

INTERVIEW PREPARATION GUIDE

Java Backend Developer Infosys

Core Java · Spring Boot · Microservices · Kafka · SQL · HR
Prepared for Neelima Jana · Interview: 5 September

How to use this guide. Work the sections in order. The plan on the next page is timed — if you fall behind, drop from the bottom, never from the top.

● **Red dot** = high priority, almost certainly asked. ●● = your differentiator. ● **Amber** = worth knowing.

✓ = self-check before moving on. ⚠ = common trap or something to verify.

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Infosys — Java Backend Developer Interview Prep

For Neelima Jana · Interview: 5 September · Plan starts 2 September

How this guide is written

Every topic follows the same four steps, so you always know what you are reading:

1. **The context** — what the thing is and what problem it solves, in plain words.
2. **A simple example** — small code you can actually follow.
3. **Why it matters** — where it is used in real work.
4. **"Say this"** — the answer to give in the room.

Do not memorise the answers. Understand the idea, so that when the question comes out slightly differently, you can still answer it.

1. What the job description is really telling you

The posting is an Infosys "Infoscion" template, so most of it describes how Infosys works rather than the role itself. The part that actually decides your interview is near the bottom:

Primary skills: Java → Spring Boot. Technology → Microservices.
Preferred: Microservices API Management, Java → Spring Boot.

That is the whole technical brief. **Java, Spring Boot and Microservices.** Files 01 to 06 cover those, and they are where you should spend most of your three days.

But do not skip the rest of the posting, because it tells you what the **managerial round** will be about:

The JD says	What it means for you
"problem definition, effort estimation, diagnosis, solution generation"	They want to see structured thinking, not just coding. File 10.
"research... literature surveys... vendor evaluation... build POCs"	Learning agility. Be ready for "how do you learn a new technology?"
"create requirement specifications... define to-be processes"	You will be asked how you handle unclear requirements. File 12.
"diagnose the root cause of issues, seek clarifications"	Your production-incident story is directly relevant. File 11.

The JD says	What it means for you
"work with clients... refining, analyzing, structuring relevant data"	Client communication. This is where your recruitment background helps you.
"assess current processes, identify improvement areas"	Your two performance-optimisation stories are exactly this.
"one or two industry domain knowledge"	You have banking . Say so early and often.

The two things that make you a strong fit

1. You have real performance numbers. 15 seconds to 6 seconds. 2-3 seconds to under 1 second. Most candidates at this level say "I optimised the API." You can say what was wrong (N+1 queries), what you did (indexing, lazy loading, refactored JOINS), and what changed. That is rare, and Infosys interviewers dig for exactly this.

2. You have banking domain experience. The JD explicitly asks for "one or two industry domain knowledge". You worked on a UK private bank's transaction recall and card management systems. Infosys staffs by domain, so this makes you easier to place, and easier to place means easier to hire.

2. What the interview will look like

Infosys typically runs **two or three rounds**, often on the same day.

Round 1 — Technical (45-60 min).

Core Java, Collections, Java 8, Spring Boot, Hibernate/JPA, SQL, and usually one coding question. Expect rapid-fire questions rather than long discussions.
→ Files 01, 02, 03, 04, 08, 09.

Round 2 — Technical / Architecture (30-45 min).

Microservices, messaging, security, design questions, and a deep dive into your projects. This is where your Kafka and RabbitMQ work stands out.
→ Files 05, 06, 07, 10, 11.

Round 3 — Managerial + HR (20-30 min).

Behavioural questions, the JD's consulting language, your career change, notice period, location, salary.
→ File 12.

A note on Infosys specifically. They are a large services company, so the technical bar is "solid and reliable" rather than "brilliant". They screen hard for **fundamentals** (Core Java and Collections especially), **communication**, and **whether you will be easy to deploy to a client**. Depth on HashMap internals will serve you better here than a clever algorithm.

3. The three-day plan

Day 1 — Tuesday 2 September (today, from now)

Goal: the fundamentals they screen hardest on.

Block	Topic	File
0:00 – 0:15	Read this file	README.md
0:15 – 2:00	Core Java and Collections ●●	01-core-java.md
2:00 – 3:00	Java 8 — streams, lambdas, Optional ●	02-java8-functional.md
3:00 – 3:15	Break	—
3:15 – 4:45	Spring Boot and dependency injection ●●	03-spring-boot.md
4:45 – 6:00	JPA, Hibernate and the N+1 problem ●●	04-jpa-hibernate.md
Evening	Say your two performance stories out loud	11-your-experience.md \$2

Day 2 — Wednesday 3 September

Goal: the primary skill on the JD, plus your differentiators.

Block	Topic	File
0:00 – 1:30	Microservices ●●	05-microservices.md
1:30 – 2:30	Kafka and RabbitMQ ●	06-kafka-rabbitmq.md
2:30 – 3:15	Spring Security and JWT ●	07-spring-security-jwt.md
3:15 – 4:30	SQL and query optimisation ●	08-sql-and-databases.md
4:30 – 6:00	Coding round practice — write the code, do not read it	09-coding-round.md
Evening	Your experience and the career change ●	11-your-experience.md

Day 3 — Thursday 4 September

Goal: consolidate, and prepare the non-technical half.

Block	Topic	File
0:00 – 1:15	Backend system design ●	10-system-design.md
1:15 – 2:15	HR and managerial round ●	12-hr-and-behavioural.md
2:15 – 3:30	Re-do the coding problems you got wrong	09-coding-round.md
3:30 – 4:30	Weak areas from days 1 and 2 — your call	—
4:30 – 5:00	Say your intro and project pitches out loud, twice	11-your-experience.md

Block	Topic	File
By 11:00 PM	Sleep.	—

Day 4 — Friday 5 September, interview day

Time	What
Wake, then 45 min	Cheat sheet only — 13-final-cheatsheet.md
30 min	Say your intro, the career-change answer and both project stories out loud
20 min	Re-read the questions you will ask them
20 min	Laptop, charger, network, ID, printed resumes, water
Before	Close everything. Breathe.

Learn nothing new on interview day. New material only pushes out what has already settled and raises your anxiety. Revision only.

4. The files

File	Topic	Priority
01-core-java.md	OOP, Collections, HashMap internals, String, exceptions	● Highest
02-java8-functional.md	Streams, lambdas, functional interfaces, Optional	● High
03-spring-boot.md	IoC, DI, annotations, bean scopes, auto-configuration	● Highest
04-jpa-hibernate.md	JPA, lazy vs eager, N+1 , transactions, caching	● Highest
05-microservices.md	Patterns, Eureka, API Gateway, resilience, saga	● Highest
06-kafka-rabbitmq.md	Messaging, when to use which — your differentiator	● High
07-spring-security-jwt.md	Authentication, RBAC, JWT flow	● Medium
08-sql-and-databases.md	Joins, indexing, execution plans, optimisation	● High
09-coding-round.md	Coding problems with full solutions	● High
10-system-design.md	Backend design questions and a framework	● Medium
11-your-experience.md	Your stories, and the two hard resume questions	● Highest
12-hr-and-behavioural.md	Infosys HR, the career change, salary, questions to ask	● High
13-final-cheatsheet.md	One page. Interview morning only.	● Read last

PDF versions are in [pdf/](#). The cheat sheet is meant to be printed.

5. Three rules for the room

1. Never say "I don't know" and stop. Say instead:

"I have not used that directly. My understanding is that it does X — is that the direction you mean?"

Service company interviewers are checking whether you can reason and whether you will admit a gap honestly. Both matter more than knowing everything.

2. Always give the *why*, not just the *what*.

"HashMap stores key-value pairs" is a textbook answer. "HashMap gives O(1) lookup because it hashes the key to a bucket index instead of scanning" is an engineer's answer. Same length, completely different impression.

3. Bring every answer back to something you actually did.

You have banking projects with real numbers. When you explain the N+1 problem, finish with "that is exactly what took our report generation from 15 seconds to 6." That is the sentence they remember.

6. Two things on your resume to sort out before the 5th

Read **file 11, section 5** properly. There are two items that a careful interviewer will ask about, and Infosys runs formal background verification, so both need a clear and honest answer ready:

1. Your summary says **"2+ years"**, but Dec 2024 to Sept 2026 is about **21 months**.
2. The move from **Talent Acquisition Executive** to **Software Engineer**.

Neither is a problem. Both are a problem if you are caught unprepared by them.

01 · Core Java and Collections

Time needed: 105 minutes

This is where Infosys screens hardest. Service company interviews go deep on fundamentals, because fundamentals are what transfer across the many different client projects they will put you on. Expect rapid-fire questions here.

If you only master one topic in this file, make it HashMap. It is the single most asked Java interview question in India.

Part A · OOP — the four pillars

You will be asked to name them and give an example of each. Have a real example ready, not a "Dog extends Animal" one.

| Q1. Encapsulation

What it is: keeping data private and exposing it only through methods.

```
public class Account {  
    private double balance;                // nobody can touch this directly  
  
    public void deposit(double amount) {  
        if (amount <= 0) throw new IllegalArgumentException("Amount must be positive");  
        this.balance += amount;           // the rule is enforced here  
    }  
  
    public double getBalance() { return balance; }  
}
```

Why it matters: if `balance` were public, any code anywhere could set it to a negative number and no validation would run. Encapsulation means the object protects its own rules. In a banking system that is not a style preference.

| Q2. Inheritance

One class reuses another's behaviour with `extends`. A `SavingsAccount extends Account` gets `deposit()` for free and adds interest calculation.

The follow-up: "Why does Java not support multiple inheritance of classes?"

"Because of the diamond problem. If B and C both extend A and override the same method, and D extends both, the compiler cannot decide which version D inherits. Java avoids that by allowing only one superclass. You can implement multiple interfaces, and since Java 8 interfaces can have default methods — so if two interfaces give the same default method, the compiler forces you to override it and resolve the conflict explicitly."

Q3. Polymorphism •

"One interface, many implementations." Two kinds, and interviewers want both.

	Overloading (compile-time)	Overriding (runtime)
What changes	Same method name, different parameters	Same signature, different class
Decided	At compile time, by the compiler	At runtime, by the actual object
Also called	Static polymorphism	Dynamic polymorphism

```
// Overloading - same name, different parameter lists
void notify(String email) { }
void notify(String email, String sms) { }

// Overriding - the subclass replaces the parent's behaviour
class Notifier { void send() { System.out.println("generic"); } }
class EmailNotifier extends Notifier { @Override void send() { System.out.println("email"); } }

Notifier n = new EmailNotifier();
n.send(); // "email" - decided by the OBJECT, not the reference type
```

That last line is the whole point of runtime polymorphism, and it is what makes Spring's dependency injection work: the field is declared as an interface, and the concrete implementation is chosen at runtime.

Real-time example. In the notification flow, the service holds a `List<Notifier>` containing an email notifier and an SMS notifier. It calls `notifier.send(...)` on each without knowing or caring which is which. When a push notifier is added later, the sending code does not change at all — that is polymorphism doing real work.

Trap: "Can you override a static method?" → No. Static methods belong to the class, not the instance, so they are **hidden**, not overridden. Which one runs is decided by the reference type, not the object.

Q4. Abstraction

Hiding *how* something works and exposing *what* it does.

● Abstract class vs interface — near-certain question

	Abstract class	Interface
Fields	Can have normal instance fields	Only <code>public static final</code> constants
Constructor	Yes	No
Methods	Abstract and concrete	Abstract, plus <code>default</code> and <code>static</code> since Java 8
Inheritance	One only	Many
Access modifiers	Any	Methods are public

When to use which — say it this way:

"An abstract class is for an 'is-a' relationship where the subclasses share state and some common implementation — a base `Transaction` class holding amount and timestamp. An interface is a contract about capability with no shared state, like `Auditable` or `PaymentProcessor`. Since I can implement many interfaces but extend only one class, I default to interfaces and use an abstract class when there is real shared code and fields."

Real-time example — say this if they ask for one.

"In the card system, `PaymentProcessor` was an **interface** with implementations for different rails — NEFT, RTGS, IMPS. Each one processes a payment completely differently, so there is no shared code to inherit, only a shared contract. The service just depends on `PaymentProcessor` and Spring injects the right one.

`BaseTransaction` was an **abstract class**, because every transaction genuinely has an amount, a currency, a timestamp and an audit trail, plus a common `validate()` that all of them run. That is shared state and shared code, which an interface cannot give me."

Part B · Collections ●●

Q5. The map of the framework

Collection (interface)

└ List	– ordered, allows duplicates	→ ArrayList, LinkedList, Vector
└ Set	– no duplicates	→ HashSet, LinkedHashSet, TreeSet
└ Queue	– FIFO processing	→ PriorityQueue, ArrayDeque

Map (separate, NOT a Collection) – key/value → HashMap, LinkedHashMap, TreeMap, Hashtable

⚠ `Map` does **not** extend `Collection`. It stores pairs, not single elements. A small detail interviewers like to check.

Q6. ArrayList vs LinkedList ●

	ArrayList	LinkedList
Structure	Growable array	Doubly linked list
Get by index	O(1)	O(n) — must walk the chain
Add/remove at end	O(1) amortised	O(1)
Add/remove in middle	O(n) — shifts elements	O(1) <i>once you are at the node</i>
Memory	Less	More — each node stores two pointers

Say this:

"I use ArrayList by default. Random access is O(1) and the elements sit together in memory, which is cache friendly. LinkedList only wins when I am inserting or removing frequently in the middle while already holding a reference to that position. In practice that is rare, and even for a queue I would use ArrayDeque."

Real-time example. A recall report loads a page of transactions and then iterates them to build the output — read-heavy, index access, no insertion in the middle. That is an `ArrayList`. I have not had a case in these systems where `LinkedList` was the right answer.

How does ArrayList grow? It starts at capacity 10, and when full it creates a new array about **1.5 times** the size and copies everything across. That copy is why `new ArrayList<>(expectedSize)` matters when you know the size up front.

Q7. ●● How does HashMap work internally?

Learn this properly. It is the most asked Java question in Indian interviews.

Storing a value — `map.put(key, value)`

1. Java calls `key.hashCode()` to get an int.
2. HashMap applies its own **hash spreading** on top: `h ^ (h >>> 16)`. This mixes the high bits into the low bits, because the next step only uses the low bits.
3. It finds the bucket: `index = (n - 1) & hash`, where `n` is the array length. Since `n` is always a power of two, this is a fast bitwise AND instead of a modulus.
4. If the bucket is empty, it stores the entry there.
5. If the bucket already has entries — a **collision** — it walks them comparing with `equals()`. Same key means replace the value; otherwise append the new entry.

The structure

Internally it is an **array of buckets**, and each bucket holds a **linked list** of entries. Since **Java 8**, once a single bucket exceeds **8 entries** (and the table is at least 64 long) that list is converted into a **balanced red-black tree**, turning worst-case lookup from $O(n)$ into $O(\log n)$. Below 6 entries it converts back.

Resizing

Default capacity **16**, load factor **0.75**. When size passes $16 \times 0.75 = 12$, the array doubles to 32 and everything is **rehashed** into new buckets. Resizing is expensive, which is why you pre-size a map you know will be large.

Complexity

$O(1)$ average for get and put. $O(\log n)$ worst case since Java 8, thanks to treeification.

Real-time example — where you actually used one

"Two places. In the report generation I loaded the reference data once — branch codes and status descriptions — into a **HashMap** keyed by code, so the loop could look each one up in $O(1)$ instead of hitting the database per row. And on the Kafka consumer side, a map of processed message IDs is how you make a consumer idempotent: check the map before acting, so a redelivered message is a no-op."

• The follow-up that catches people: equals and hashCode

"If you use a custom object as a key you must override both **hashCode()** and **equals()**. The contract is that **equal objects must have equal hash codes**. If I override **equals** but not **hashCode**, two objects that are equal can produce different hash codes, land in different buckets, and the map will never find the entry I stored. That is also why **HashMap** keys should be **immutable** — if a field used in the hash changes after insertion, the entry is effectively lost, because it is sitting in a bucket that no longer matches its hash."

Q8. HashMap vs Hashtable vs ConcurrentHashMap •

	HashMap	Hashtable	ConcurrentHashMap
Thread safe	No	Yes	Yes
How	—	Locks the whole map	Locks only the affected bucket
Null key/value	One null key, many null values	Neither	Neither
Performance	Fastest single-threaded	Slow — everything serialises	Fast under concurrency

	HashMap	Hashtable	ConcurrentHashMap
Status	Standard	Legacy, do not use	Standard for concurrent use

"Hashtable synchronises every method, so all threads queue behind one lock. That is why it is effectively obsolete. ConcurrentHashMap locks at bucket level, so threads writing to different buckets do not block each other, which is why it scales."

Real-time example. A Spring `@Service` is a **singleton**, so any collection held in a field is shared by every request thread at once. If I keep an in-memory counter or cache in a service — say a per-branch request count — a plain `HashMap` will corrupt under load and can even spin forever on an older JDK. That field has to be a `ConcurrentHashMap`.

Q9. HashSet vs LinkedHashSet vs TreeSet

	Ordering	Backing structure	Complexity
HashSet	None guaranteed	HashMap	O(1)
LinkedHashSet	Insertion order	HashMap + linked list	O(1)
TreeSet	Sorted	Red-black tree	O(log n)

Useful detail: `HashSet` is literally a `HashMap` where every value is the same dummy object. That is why its rules about `hashCode` and `equals` are identical.

Real-time example. `HashSet` for "which account numbers have already been processed in this batch" — I only care about membership. `TreeSet` when the order matters, like interest rate slabs that must be walked from lowest to highest to find the applicable band. `LinkedHashSet` when I need uniqueness but the output must stay in the order the records arrived, which matters for a report a human reads.

Q10. Fail-fast vs fail-safe iterators

"Fail-fast iterators, like `ArrayList`'s and `HashMap`'s, throw `ConcurrentModificationException` if the collection is structurally modified while iterating. They track a `modCount` and check it on each `next()`. Fail-safe iterators, like `ConcurrentHashMap`'s or `CopyOnWriteArrayList`'s, work on a snapshot or allow concurrent access, so they do not throw — but they may not reflect the very latest changes."

How do you remove safely while iterating? Use `iterator.remove()`, or `collection.removeIf(predicate)`. Never `list.remove()` inside a for-each loop.

Q11. Comparable vs Comparator ●

```
// Comparable - the class defines its OWN natural ordering. One only.
class Txn implements Comparable<Txn> {
    public int compareTo(Txn other) { return this.date.compareTo(other.date); }
}

// Comparator - ordering defined OUTSIDE the class. As many as you like.
Comparator<Txn> byAmount = Comparator.comparing(Txn::getAmount);
list.sort(byAmount.reversed());
```

"Comparable is the natural order and lives inside the class, so there can be only one. Comparator lives outside, so I can have several — sort transactions by date on one screen and by amount on another without touching the entity."

Real-time example. `Transaction` implements `Comparable` on `createdAt`, because newest-first is its natural order everywhere in the application. But the recall report screen lets the user sort by amount, by status or by customer name — three `Comparator`s, none of which required changing the entity.

Part C · Strings, immutability and memory

Q12. Why is String immutable? ●

Once created, a String's value can never change. `s.concat("x")` returns a **new** String.

The four reasons — give at least two:

1. **String pool.** Java reuses identical literals to save memory. That is only safe if nobody can change them underneath other references.
2. **Security.** Usernames, file paths and DB URLs are Strings. If they were mutable, a value could be validated and then changed before use.
3. **Thread safety.** Immutable objects can be shared across threads with no locking.
4. **Hashcode caching.** String caches its hash, which makes it a fast HashMap key. That is only valid if the value cannot change.

Real-time example. Account numbers and transaction reference numbers are Strings used as map keys and passed between services. If a String were mutable, code holding a reference could change an account number *after* it had been validated and after it had been used to place an entry in a map — the entry would then be unreachable. Immutability is what makes them safe to share.

Q13. String vs StringBuilder vs StringBuffer

	Mutable?	Thread safe	Use when
String	No	Yes (immutable)	Fixed values
StringBuilder	Yes	No	Building strings — the default
StringBuffer	Yes	Yes (synchronised)	Legacy; rarely needed

"Concatenating in a loop with `+` creates a new String every iteration, so a thousand-iteration loop creates a thousand throwaway objects. StringBuilder mutates one buffer, so it is $O(n)$ instead of $O(n^2)$."

Q14. `==` vs `.equals()`

`==` compares **references** — are these the same object in memory.

`.equals()` compares **values** — as defined by the class.

```
String a = "hello";           // goes into the string pool
String b = "hello";           // reuses the SAME pooled object
String c = new String("hello"); // forces a NEW object on the heap

a == b      // true  - same pooled reference
a == c      // false - different objects
a.equals(c) // true  - same characters
```

This is a favourite quick question. Learn the three lines.

Q15. Stack vs Heap

- **Stack** — one per thread. Holds method calls, local variables and references. Freed automatically when the method returns.
- **Heap** — shared by all threads. Holds all objects. Managed by the garbage collector.

`StackOverflowError` means runaway recursion. `OutOfMemoryError` means the heap is full, usually a leak or an unbounded collection.

Part D · Exceptions

Q16. Checked vs unchecked •

```
Throwable
├── Error                - JVM level, do not catch (OutOfMemoryError)
└── Exception
    ├── RuntimeException - UNCHECKED (NullPointerException, IllegalArgumentException)
    └── everything else - CHECKED (IOException, SQLException)
```

- **Checked:** the compiler forces you to catch it or declare `throws`. Represents recoverable, expected conditions — a missing file, a network failure.
- **Unchecked:** extends `RuntimeException`. Usually a programming bug. No compiler enforcement.

Say this:

"In application code I mostly use unchecked exceptions and a custom hierarchy — a `TransactionNotFoundException` extending `RuntimeException` — because forcing every caller to catch and rethrow adds noise. I handle them centrally in Spring with `@ControllerAdvice` so the API returns a consistent error shape. It also matters for `@Transactional`, because by default Spring only rolls back on unchecked exceptions."

That last sentence connects two topics and sounds like real experience. Use it.

Real-time example. In the recall APIs I had `TransactionNotFoundException` and `InvalidRecallStateException`, both unchecked, both extending a common `BusinessException`. `@RestControllerAdvice` mapped them to 404 and 409 respectively, so the UI team always got the same error shape. Making them unchecked also mattered for `@Transactional`, because a checked exception would have **committed** the transaction instead of rolling it back.

Q17. try-with-resources and finally

```
try (Connection conn = dataSource.getConnection()) {
    // conn.close() is called automatically, even if an exception is thrown
}
```

Anything implementing `AutoCloseable` works. This replaced the old `finally { if (conn != null) conn.close(); }` pattern.

Trap: "Does `finally` always run?" → Almost always. Not if the JVM exits (`System.exit()`) or the thread is killed. And if `finally` contains a `return`, it overrides the one in `try` — which is why you never return from `finally`.

Q18. `throw` VS `throws` VS `final` VS `finally` VS `finalize`

- `throw` — actually raise an exception now.
- `throws` — declare that this method may raise one.

- `final` — a constant, a non-overridable method, or a non-extendable class.
 - `finally` — the block that runs whatever happens.
 - `finalize()` — a deprecated, unreliable method the GC used to call. Never use it.
-

✓ Check yourself before moving on

1. Explain **HashMap put and get end to end** — hashing, buckets, collisions, treeification, resizing. Out loud, without notes.
2. Explain the `equals` / `hashCode` contract and what breaks if you ignore it.
3. Overloading vs overriding, with an example of each.
4. Abstract class vs interface, and when you would pick each.
5. Explain the three lines of the `==` vs `.equals()` String example.

02 · Java 8 and Functional Programming

Time needed: 60 minutes

Your resume says "Java 8+", so this **will** be tested. Interviewers usually ask you to explain streams and then write one on the spot.

First, the context: what problem did Java 8 solve?

Before Java 8, processing a collection meant writing *how* to loop:

```
// imperative - you describe every step
List<String> names = new ArrayList<>();
for (Txn t : txns) {
    if (t.getAmount() > 1000) {
        names.add(t.getCustomerName().toUpperCase());
    }
}
Collections.sort(names);
```

Java 8 let you describe *what* you want instead:

```
// declarative - the intent is visible in one read
List<String> names = txns.stream()
    .filter(t -> t.getAmount() > 1000)
    .map(t -> t.getCustomerName().toUpperCase())
    .sorted()
    .collect(Collectors.toList());
```

Same result, but the second reads as a sentence: filter, map, sort, collect. That is the point — less boilerplate and clearer intent.

Q1. What is a lambda? ●

A short way to write an implementation of an interface that has exactly one method.

```
// before Java 8
Runnable r = new Runnable() {
    public void run() { System.out.println("running"); }
};

// with a lambda
Runnable r = () -> System.out.println("running");
```

The compiler knows `Runnable` has one method, so you only supply the body.

Q2. What is a functional interface? •

An interface with **exactly one abstract method**. That is what makes it a valid target for a lambda. `@FunctionalInterface` is optional but makes the compiler enforce it.

The four you must know:

Interface	Method	Takes	Returns	Used by
<code>Predicate<T></code>	<code>test(T)</code>	T	boolean	<code>filter</code>
<code>Function<T,R></code>	<code>apply(T)</code>	T	R	<code>map</code>
<code>Consumer<T></code>	<code>accept(T)</code>	T	void	<code>forEach</code>
<code>Supplier<T></code>	<code>get()</code>	nothing	T	<code>orElseGet</code> , lazy values

```
Predicate<Txn> isLarge = t -> t.getAmount() > 1000;
Function<Txn, String> name = t -> t.getCustomerName();
Consumer<Txn> log = t -> System.out.println(t);
Supplier<Txn> empty = () -> new Txn();
```

Also worth naming: `BiFunction<T,U,R>` (two arguments), `UnaryOperator<T>` (T to T), and `BinaryOperator<T>` (two Ts to one T, used by `reduce`).

Q3. Method references

Shorthand for a lambda that just calls one method.

```
.map(t -> t.getCustomerName()) → .map(Txn::getCustomerName)
.forEach(x -> System.out.println(x)) → .forEach(System.out::println)
```

Q4. •• Streams — intermediate vs terminal operations

This distinction is asked constantly.

Intermediate operations return another Stream and are **lazy** — they do nothing until a terminal operation runs. `filter`, `map`, `flatMap`, `sorted`, `distinct`, `limit`, `skip`, `peek`.

Terminal operations produce a result and **trigger** the whole pipeline. `collect`, `forEach`, `reduce`, `count`, `min`, `max`, `anyMatch`, `allMatch`, `findFirst`.

```
txns.stream().filter(t -> t.getAmount() > 1000); // NOTHING happens - no terminal op
```

Why laziness matters — the answer that impresses:

"Because intermediate operations are lazy, the stream can fuse them into a single pass and stop early. `stream().filter(...).findFirst()` does not filter the whole list and then take the first — it stops at the first match. With `limit(10)` on a million records, only enough elements are processed to produce ten."

A stream can only be consumed once. Reusing one throws

`IllegalStateException: stream has already been operated upon or closed`.

Q5. The stream methods you must be able to write ●

```
// filter + map + collect
List<String> big = txns.stream()
    .filter(t -> t.getAmount() > 1000)
    .map(Txn::getCustomerName)
    .collect(Collectors.toList());

// sum with reduce, and the shorter specialised form
double total = txns.stream().mapToDouble(Txn::getAmount).sum();
double same = txns.stream().map(Txn::getAmount).reduce(0.0, Double::sum);

// GROUP BY - very commonly asked
Map<String, List<Txn>> byStatus =
    txns.stream().collect(Collectors.groupingBy(Txn::getStatus));

// group and count
Map<String, Long> countByStatus =
    txns.stream().collect(Collectors.groupingBy(Txn::getStatus, Collectors.counting()));

// group and sum
Map<String, Double> totalByStatus = txns.stream()
    .collect(Collectors.groupingBy(Txn::getStatus,
        Collectors.summingDouble(Txn::getAmount)));

// sort by a field, descending
txns.stream().sorted(Comparator.comparing(Txn::getAmount).reversed())

// flatMap - flatten nested collections into one stream
List<Item> all = orders.stream()
    .flatMap(o -> o.getItems().stream())
    .collect(Collectors.toList());
```

Real-time example — where this replaced real code. The report summary needed totals per status: how many recalls are pending, approved and rejected, and the total value of each. That was originally nested loops with counters and a map being built by hand. One

`groupingBy` with `summingDouble` replaced about twenty lines, and the intent is now visible in one read.

⚠ But note the boundary: **do this in SQL when the data is in the database**. Streaming a million rows into memory to group them is much slower than a `GROUP BY`. Streams are for data you already have in hand.

That caveat is worth saying out loud — it shows you know where the work should happen.

● `map` vs `flatMap` is a guaranteed question:

"`map` transforms one element into one element. `flatMap` transforms one element into a **stream** of elements and then flattens them all into a single stream. If each order has a list of items, `map` gives me a stream of lists, and `flatMap` gives me a stream of items."

Real-time example. A recall request has a list of approvals. To get every approver across a page of requests, `map` would give me a stream of *lists* of approvals; `flatMap` gives me a single stream of approvals I can then filter and count.

Q6. Optional ●

Introduced to make "this might be absent" explicit in the type, instead of returning `null` and hoping the caller checks.

```
Optional<Txn> found = repository.findById(id);

// good
String name = found.map(Txn::getCustomerName).orElse("Unknown");
found.ifPresent(t -> log.info("Found {} ", t.getId()));
Txn txn = found.orElseThrow(() -> new TransactionNotFoundException(id));

// bad - this is just a null check with extra steps
if (found.isPresent()) { Txn t = found.get(); }
```

"Optional makes the possibility of absence part of the method signature, so the caller cannot ignore it by accident. I use `map`, `orElse` and `orElseThrow` rather than `isPresent` and `get`, because that defeats the purpose. Spring Data returns `Optional` from `findById` for exactly this reason."

Real-time example. `txnRepository.findById(id)` returns `Optional<Transaction>`. In the recall service that becomes `.orElseThrow(() -> new TransactionNotFoundException(id))`, which the `@RestControllerAdvice` turns into a clean 404. Before `Optional`, that same code path returned `null` and produced a `NullPointerException` and a 500 — which tells the caller nothing useful.

orElse vs **orElseGet** — a good detail: `orElse(buildDefault())` evaluates the argument **always**, even when a value is present. `orElseGet(() -> buildDefault())` only runs it when needed. It matters when the default is expensive.

Q7. Parallel streams — and when not to use them

```
list.parallelStream().filter(...).collect(...);
```

"It splits the work across the fork-join pool. It only helps with a large dataset and a genuinely CPU-bound operation with no shared mutable state. For small collections the coordination overhead makes it slower, and for I/O-bound work like database calls it is the wrong tool. I would measure before using it, not assume."

Q8. Other Java 8 features worth naming

- **Default and static methods in interfaces** — let you add a method to an interface without breaking every existing implementation.
- **java.time** — `LocalDate`, `LocalDateTime`, `Duration`. Immutable and thread safe, unlike the old `Date` and `Calendar`.
- **CompletableFuture** — asynchronous composition, useful for calling several services in parallel.

Q9. Later versions, in one line each

If asked "what have you used beyond Java 8":

- **Java 11** — `var` for local variables, `String.isBlank()`, `strip()`, `lines()`.
- **Java 17 (LTS)** — records, sealed classes, switch expressions, text blocks.
- A **record** is a compact immutable data carrier: `record TxnDto(String id, double amount) {}` generates the constructor, getters, `equals`, `hashCode` and `toString`. Very useful for DTOs.

✓ Check yourself before moving on

1. Write a stream that groups transactions by status and sums the amount in each group. From memory.
2. Explain intermediate vs terminal, and why laziness lets a stream stop early.
3. Explain `map` vs `flatMap` in two sentences.
4. Name the four core functional interfaces and which stream method uses each.

03 · Spring Boot

Time needed: 90 minutes

"Java → Spring Boot" is listed as the **primary skill** on the job description. Along with Core Java, this is where most of your technical round will be spent.

| First, the context: what problem does Spring solve?

The problem: tight coupling

Without Spring, an object creates the things it depends on:

```
public class TransactionService {  
    private final EmailNotifier notifier = new EmailNotifier();    // hard-wired  
}
```

Two problems. To switch to SMS you must edit this class. And in a unit test you cannot replace the notifier with a fake, so your test sends real emails.

The solution: Inversion of Control

The object stops creating its dependencies and **receives** them instead:

```
@Service  
public class TransactionService {  
    private final Notifier notifier;    // an interface  
  
    public TransactionService(Notifier notifier) {    // handed in from outside  
        this.notifier = notifier;  
    }  
}
```

Now the class depends on an interface, not a concrete class. Something else decides which implementation to supply. That "something else" is the Spring container.

Inversion of Control (IoC) is the principle: control of creating objects is inverted, moving from your class to the framework.

Dependency Injection (DI) is how Spring implements it: it constructs objects and passes their dependencies in.

Say it in one line:

"IoC is the principle that the framework, not my class, controls object creation and wiring. Dependency injection is the mechanism. The practical benefit is that my classes depend on interfaces, so I can swap implementations and inject mocks in tests without changing the class."

Real-time example. The card approval service depends on a `NotificationSender` interface. In production Spring injects the RabbitMQ implementation. In unit tests I pass a mock and assert that approval publishes exactly one message — no broker, no emails, and the test runs in milliseconds. That is only possible because the service never says `new`.

And Spring Boot on top of Spring

Plain Spring needed a lot of XML and manual configuration. Spring Boot adds:

- **Auto-configuration** — sees a database driver on the classpath and configures a `DataSource` for you.
- **Starter dependencies** — `spring-boot-starter-web` pulls in a compatible set of libraries so you are not resolving versions by hand.
- **Embedded server** — Tomcat is inside the JAR, so `java -jar app.jar` runs it. No separate application server to install.
- **Production features** — Actuator gives health checks and metrics endpoints.

"Spring Boot is Spring with opinionated defaults. It removes the configuration work so you start with a running application and override only what you need."

Q1. • The three types of injection, and which to use

```
// 1. CONSTRUCTOR injection - what you should use
@Service
public class TxnService {
    private final TxnRepository repo;
    public TxnService(TxnRepository repo) { this.repo = repo; }
}

// 2. FIELD injection - convenient, but avoid
@Autowired private TxnRepository repo;

// 3. SETTER injection - for genuinely optional dependencies
@Autowired public void setRepo(TxnRepository repo) { this.repo = repo; }
```

Why constructor injection — four reasons, give two or three:

1. The field can be `final`, so the object is **immutable** once built.
2. Dependencies **cannot be missing** — the object cannot be constructed without them.
3. **Testable without Spring** — `new TxnService(mockRepo)` just works. With field injection you need reflection or a Spring context.

4. It makes bad design **visible** — a constructor with eight parameters is obviously doing too much. Field injection hides that.

Since Spring 4.3, if a class has one constructor you do not even need `@Autowired`.

Q2. • The annotations you must know

Annotation	What it does
<code>@SpringBootApplication</code>	The main class. Combines the next three.
<code>@Configuration</code>	This class defines beans
<code>@EnableAutoConfiguration</code>	Turn on Boot's auto-config
<code>@ComponentScan</code>	Scan this package and below for components
<code>@Component</code>	A generic Spring-managed bean
<code>@Service</code>	A <code>@Component</code> marking business logic
<code>@Repository</code>	A <code>@Component</code> for data access — also translates DB exceptions into Spring's <code>DataAccessException</code> hierarchy
<code>@Controller</code>	Returns view names
<code>@RestController</code>	<code>@Controller</code> + <code>@ResponseBody</code> — returns JSON
<code>@Autowired</code>	Inject a dependency
<code>@Qualifier("name")</code>	Choose between multiple beans of the same type
<code>@Primary</code>	Make one bean the default when several match
<code>@Value("\${prop}")</code>	Inject a property from configuration
<code>@Bean</code>	Declare a bean from a method, usually for third-party classes
<code>@Transactional</code>	Wrap the method in a database transaction
<code>@ControllerAdvice</code>	Global exception handling across controllers

Trap: '`@Component` vs `@Service` vs `@Repository` — is there a functional difference?'

"Technically `@Service` and `@Component` behave identically; the difference is intent, so the layer is obvious to a reader and to tooling. `@Repository` is the exception — it does real work, translating vendor-specific database exceptions into Spring's `DataAccessException` hierarchy, so my service layer is not coupled to a particular database's error codes."

Q3. REST annotations

```
@RestController
@RequestMapping("/api/transactions")
public class TxnController {

    @GetMapping("/{id}")
    public TxnDto get(@PathVariable Long id) { ... }

    @GetMapping
    public List<TxnDto> search(@RequestParam(defaultValue = "0") int page) { ... }

    @PostMapping
    @ResponseStatus(HttpStatus.CREATED)
    public TxnDto create(@Valid @RequestBody CreateTxnRequest req) { ... }

    @PutMapping("/{id}")
    public TxnDto replace(@PathVariable Long id, @RequestBody TxnDto dto) { ... }

    @DeleteMapping("/{id}")
    public ResponseEntity<Void> delete(@PathVariable Long id) { ... }
}
```

`@PathVariable` reads from the URL path. `@RequestParam` reads a query parameter.
`@RequestBody` deserialises the JSON body. `@Valid` triggers Bean Validation.

Q4. • Bean scopes

Scope	Meaning
singleton	The default. One instance for the whole application context.
prototype	A new instance every time it is requested.
request	One per HTTP request (web only).
session	One per HTTP session (web only).

• **The follow-up:** *"Singleton beans are shared across all requests. Is that a problem?"*

"Only if the bean holds mutable state. Spring beans should be stateless — request data belongs in method parameters and local variables, not in instance fields. A singleton service with a mutable instance field is a genuine concurrency bug, because many threads share that one object."

Real-time example — a real bug this causes. If a service stores the current request's user in an instance field to avoid passing it around, two concurrent approvals will

overwrite each other's value, and one approval gets attributed to the wrong approver. In a bank that is an audit failure, not just a bug. Request-scoped data belongs in method parameters, or in the `SecurityContext`, never in a field on a singleton.

That answer shows you understand *why* the default is safe, which is what they are checking.

Q5. • Global exception handling

Nearly always asked, because it separates people who have shipped an API from people who have followed a tutorial.

```
@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(TransactionNotFoundException.class)
    public ResponseEntity<ErrorResponse> handleNotFound(TransactionNotFoundException ex) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND)
            .body(new ErrorResponse("TXN_NOT_FOUND", ex.getMessage()));
    }

    @ExceptionHandler(MethodArgumentNotValidException.class)
    public ResponseEntity<ErrorResponse> handleValidation(MethodArgumentNotValidException ex) {
        String msg = ex.getBindingResult().getFieldErrors().stream()
            .map(e -> e.getField() + ": " + e.getDefaultMessage())
            .collect(Collectors.joining(", "));
        return ResponseEntity.badRequest().body(new ErrorResponse("VALIDATION_ERROR", msg));
    }

    @ExceptionHandler(Exception.class)
    public ResponseEntity<ErrorResponse> handleAll(Exception ex) {
        log.error("Unexpected error", ex);    // log the detail...
        return ResponseEntity.status(500)
            .body(new ErrorResponse("INTERNAL_ERROR", "Something went wrong"));
        // ...but do not leak it to the client
    }
}
```

Say why it matters:

"It keeps error handling out of every controller, so the API returns one consistent error shape. The last handler matters for security: I log the stack trace but return a generic message, because a raw stack trace tells an attacker about your framework versions and internal structure. On a banking system that is a real concern."

Real-time example. Before this was centralised, each controller had its own try/catch and every endpoint returned a slightly different error body — some with a `message`, some with an `error`, some just a stack trace. The UI team had to special-case

each one. One `@RestControllerAdvice` gave every endpoint the same `{ code, message, timestamp }` shape, and their error handling became a single function.

Q6. Bean Validation

```
public class CreateTxnRequest {
    @NotBlank(message = "Account number is required")
    private String accountNumber;

    @NotNull @DecimalMin(value = "0.01", message = "Amount must be positive")
    private BigDecimal amount;

    @Email private String notifyEmail;
}
```

`@Valid` on the `@RequestBody` triggers it, and a failure raises `MethodArgumentNotValidException`, handled above.

⚠ Note `BigDecimal` for money, never `double`. Floating point cannot represent decimal fractions exactly, so `0.1 + 0.2` is not `0.3`. In a banking interview, using `BigDecimal` for currency is a detail that gets noticed.

Q7. Configuration and profiles

```
# application.yml
spring:
  datasource:
    url: ${DB_URL}
    username: ${DB_USER}
    password: ${DB_PASSWORD}    # from environment, never committed
  ---
spring:
  config.activate.on-profile: dev
  jpa.show-sql: true
```

Run with `-Dspring.profiles.active=dev`. Profiles let one build behave differently per environment without rebuilding.

Real-time example — this is your SIT and UAT workflow. The same artefact is promoted through dev, SIT, UAT and production. Only the profile changes: different database URLs, different queue names, SQL logging on in dev and off in production. That is what makes it the *same build* being tested and released, rather than a rebuild per environment — which is the whole point of promoting a tested artefact.

```
@ConfigurationProperties(prefix = "app.recall")    // type-safe, grouped config
public class RecallProperties { private int maxDays; private String queueName; }
```

Q8. Spring Boot Actuator

Adds production endpoints: `/actuator/health` (used by load balancers and Kubernetes to decide if the instance is alive), `/actuator/metrics`, `/actuator/info`. Say that you would **secure these**, because they expose internals.

Q9. How does auto-configuration actually work?

A step above the basics, and a good one to know:

"Spring Boot ships configuration classes annotated with conditions such as `@ConditionalOnClass` and `@ConditionalOnMissingBean`. At startup it evaluates those conditions against what is on the classpath and what you have already defined. So if H2 is on the classpath and I have not declared a `DataSource`, it creates one — but the moment I define my own bean, `@ConditionalOnMissingBean` backs off and mine wins. That is why defaults never fight your own configuration."

Q10. Testing a Spring Boot application

```
@WebMvcTest(TxnController.class)           // controller layer only, fast
class TxnControllerTest {
    @Autowired MockMvc mockMvc;
    @MockBean TxnService service;           // replaces the real bean

    @Test void returns404WhenMissing() throws Exception {
        when(service.findById(1L)).thenReturn(new TransactionNotFoundException(1L));
        mockMvc.perform(get("/api/transactions/1")).andExpect(status().isNotFound());
    }
}
```

- `@SpringBootTest` loads the whole context — integration tests, slower.
- `@WebMvcTest` / `@DataJpaTest` load one slice — much faster.
- `@MockBean` swaps a bean for a Mockito mock.

Your resume mentions unit tests for state-transition validation, so be ready to say what you actually tested and why.

✓ Check yourself before moving on

1. Explain IoC and DI, and why constructor injection is preferred — three reasons.
2. Explain what a singleton bean means for thread safety.
3. Write a `@RestControllerAdvice` handler from memory.
4. Explain how auto-configuration decides what to configure.

04 · JPA, Hibernate and the N+1 Problem

Time needed: 75 minutes

This file contains your best story. Both your performance achievements — 15s → 6s and 2-3s → under 1s — came from fixing N+1 queries. When the interviewer asks about Hibernate, you are not reciting theory, you are describing something you did. Make sure you can explain the mechanism precisely enough to prove that.

| First, the context: what is JPA and what is Hibernate?

Your database stores **rows in tables**. Your Java code works with **objects**. Someone has to translate between them. Doing it by hand means endless JDBC boilerplate: open a connection, build SQL, loop the ResultSet, map each column to a field, close everything.

ORM (Object-Relational Mapping) automates that translation.

- **JPA** is the *specification* — a set of interfaces and annotations. It is a standard, not an implementation.
- **Hibernate** is the most common *implementation* of that specification.
- **Spring Data JPA** is a layer above both that generates repository implementations from method names, so you rarely write queries at all.

"JPA is the specification, Hibernate is the implementation, and Spring Data JPA is the abstraction on top that generates the repository code. Coding against JPA interfaces means I could switch the provider without rewriting my entities."

Q1. Entity mapping basics

```
@Entity
@Table(name = "transactions")
public class Transaction {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false, length = 20)
    private String accountNumber;

    private BigDecimal amount;

    @Enumerated(EnumType.STRING)    // store "PENDING", not 0 - ordinals break on reorder
    private TxnStatus status;

    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "customer_id")
    private Customer customer;
}
```

⚠ `@Enumerated(EnumType.STRING)` is worth mentioning. The default is `ORDINAL`, which stores the enum's position as a number — so inserting a new value in the middle of the enum silently corrupts every existing row. In banking data that is a serious bug.

Q2. • Lazy vs eager loading

- **EAGER** — the related entity is loaded immediately, with the parent.
- **LAZY** — a proxy is put in place, and the real data is only fetched when you first access it.

The defaults, which they may ask you to recite:

Relationship	Default
@OneToOne	EAGER
@ManyToOne	EAGER
@OneToMany	LAZY
@ManyToMany	LAZY

Say this:

"I set `@ManyToOne` to LAZY explicitly, because the default is EAGER and that means every time I load a transaction I also load its customer, even when I only wanted the amount. On a list of a thousand transactions that is a thousand unnecessary joins or queries."

Real-time example. `Transaction` has a `@ManyToOne` to `Customer` and to `Account`.

The recall list screen shows only reference number, amount and status — no customer data at all. With the EAGER default, every single row still dragged its customer and account across, which is a large part of why that page was slow. Setting them LAZY meant the list query loaded only what the screen actually displays.

The catch with LAZY: if you access a lazy field after the persistence context is closed, you get `LazyInitializationException`. The correct fixes are to fetch what you need in the query (a join fetch), or map to a DTO inside the transaction. The wrong fix is `spring.jpa.open-in-view=true`, which keeps the session open for the whole request and hides the problem while making it worse.

Q3. ●● The N+1 problem — your story

This is the most important answer in this file. Be able to explain it in 60 seconds.

What it is

```
List<Transaction> txns = txnRepository.findAll();           // 1 query - gets 100 transactions

for (Transaction t : txns) {
    System.out.println(t.getCustomer().getName());         // 1 query EACH - 100 more
}
```

One query to get the list, then N more queries — one per row — to load each row's relation. Hence **N+1**. With 100 transactions that is **101 database round trips** instead of one or two.

Why it is so slow

Each round trip has network latency. Even at 5ms each, 100 extra queries is half a second of pure waiting, and it grows linearly with the data. This is why a report that was fine in testing crawls in production — the row count went up.

How to find it

"Turn on `spring.jpa.show-sql` or set the Hibernate SQL logger to DEBUG, then load the page and count the queries. If you see the same query repeating with a different ID, that is N+1. I also looked at execution plans to confirm which queries were actually costly."

How to fix it — know all four

1. JOIN FETCH — the usual fix.

```
@Query("SELECT t FROM Transaction t JOIN FETCH t.customer WHERE t.status = :status")
List<Transaction> findByStatusWithCustomer(@Param("status") TxnStatus status);
```

One query, everything loaded.

2. @EntityGraph — the declarative version.

```
@EntityGraph(attributePaths = {"customer", "account"})
List<Transaction> findByStatus(TxnStatus status);
```

3. Batch fetching — turns N queries into N/batch queries.

```
spring.jpa.properties.hibernate.default_batch_fetch_size: 25
```

4. A DTO projection — the best option for a read-only report.

```
@Query("SELECT new com.bank.dto.TxnReportDto(t.id, t.amount, c.name) " +
        "FROM Transaction t JOIN t.customer c WHERE t.status = :status")
List<TxnReportDto> findReportData(@Param("status") TxnStatus status);
```

This selects only the columns you need, avoids loading full entities into the persistence context, and skips dirty-checking entirely.

⚠ **The trap with JOIN FETCH:** fetching **two** collections in one query produces a cartesian product. Fetch one collection at a time, or use batch fetching for the second. Mentioning this unprompted marks you out.

● Your version of the answer

"On the recall reports, generation was taking about 15 seconds. I turned on SQL logging and saw the same query repeating once per row — a classic N+1, because the report walked each transaction's related entities. I replaced it with a projection query that joined and selected only the columns the report needed, added indexes on the columns in the WHERE and JOIN clauses, and rewrote a couple of inefficient JOINS. That took it from 15 seconds to 6. On the Business Card APIs the same pattern was costing 2 to 3 seconds per call, and fixing the N+1 plus setting the right associations to lazy brought it under a second."

Practise that until it is fluent. It answers "tell me about performance tuning", "tell me about Hibernate", and "tell me about a difficult problem" all at once.

Q4. • @Transactional

```
@Transactional
public void recallTransaction(Long id) {
    Transaction txn = repo.findById(id).orElseThrow();
    txn.setStatus(RECALLED);           // no save() needed - dirty checking
    auditService.record(txn);
    // commit on normal return, rollback on a RuntimeException
}
```

The three things they ask:

1. What does it actually do? Spring creates a **proxy** around the bean. The proxy opens a transaction before the method, commits after it returns, and rolls back on failure.

2. When does it roll back? By default on **unchecked** exceptions (`RuntimeException` and `Error`) only. A checked exception **commits** unless you say `@Transactional(rollbackFor = Exception.class)`. This surprises people and is a favourite question.

3. • Why does @Transactional not work when you call the method from inside the same class?

"Because it works through a proxy. An external caller goes through the proxy, so the transaction starts. But an internal call — `this.otherMethod()` — bypasses the proxy entirely and calls the real method directly, so no transaction is started. The fix is to move that method into a separate bean, so the call goes through the proxy again."

That is a senior-level answer and interviewers remember it.

Propagation, briefly: `REQUIRED` is the default — join the existing transaction or start one. `REQUIRES_NEW` suspends the current one and starts an independent one, which is what you want for audit logging that must persist even if the main work rolls back.

Real-time example — the clearest use of `REQUIRES_NEW`. A recall attempt fails validation halfway through and the main transaction rolls back, which is correct: no partial state is written. But the **audit record of the attempt must survive**, because compliance needs to know someone tried. If the audit insert joins the main transaction it is rolled back with everything else. `REQUIRES_NEW` on the audit method gives it its own transaction that commits independently.

In a banking system that is not a nicety — an audit trail that disappears when something fails is worse than no audit trail, because you believe it.

Isolation levels, in one line each:

- `READ_UNCOMMITTED` — can see uncommitted data (dirty reads). Rarely used.
- `READ_COMMITTED` — only committed data. The usual default in PostgreSQL and Oracle.
- `REPEATABLE_READ` — the same row reads the same within a transaction. MySQL's default.
- `SERIALIZABLE` — full isolation, slowest.

Real-time example — why isolation matters here. Two approvers open the same card request and click Approve at the same moment. Both transactions read the status as PENDING, both see it as approvable, and both write APPROVED — so the request is approved twice and two approval records exist. That is a **lost update**.

The fixes are optimistic locking with an `@Version` column, so the second write fails and can be retried, or a `SELECT ... FOR UPDATE` pessimistic lock. Optimistic is usually the right choice, because genuine collisions are rare and it does not hold database locks.

Q5. Hibernate caching

- **First-level cache** — the persistence context. **Always on**, scoped to one transaction. Loading the same entity twice in one transaction hits the database once.
- **Second-level cache** — optional, shared across transactions, needs a provider like Ehcache or Hazelcast. Good for reference data that rarely changes, such as a list of branch codes. Not for data that changes often.
- **Query cache** — caches query results. Must be used with the second-level cache and is easy to get wrong.

Real-time example. Branch codes, currency codes and transaction-status descriptions are read on nearly every request and change perhaps twice a year. That is textbook second-level cache material. Transactions themselves are **never** cached that way — the data changes constantly and stale financial data is worse than a slow query.

Q6. Spring Data JPA repositories

```
public interface TxnRepository extends JpaRepository<Transaction, Long> {

    // Spring generates the query FROM THE METHOD NAME
    List<Transaction> findByStatusAndAmountGreaterThan(TxnStatus status, BigDecimal amt);

    Optional<Transaction> findByReferenceNumber(String ref);

    // your own JPQL when the name would get silly
    @Query("SELECT t FROM Transaction t WHERE t.createdAt BETWEEN :from AND :to")
    List<Transaction> findInRange(@Param("from") LocalDate from, @Param("to") LocalDate to);

    // native SQL when you need database-specific features
    @Query(value = "SELECT * FROM transactions WHERE MATCH(notes) AGAINST(:q)", nativeQuery = true)
    List<Transaction> search(@Param("q") String q);
}
```

Hierarchy: `Repository` → `CrudRepository` → `PagingAndSortingRepository` → `JpaRepository`. Each adds methods; `JpaRepository` adds JPA-specific ones like `flush()` and `saveAll()`.

Pagination, which you should mention for any list endpoint:

```
Page<Transaction> page = repo.findByStatus(PENDING, PageRequest.of(0, 20,
Sort.by("createdAt").descending()));
```

Q7. `save()` vs `saveAndFlush()`, and entity states

Three states: **transient** (a new object, not known to Hibernate), **persistent** (managed by the persistence context — changes are tracked automatically), **detached** (was persistent, but the context has closed).

Dirty checking is the thing to mention: for a managed entity you do not need to call `save()`. At flush time Hibernate compares the entity with its loaded snapshot and issues an UPDATE only for the fields that changed.

`saveAndFlush()` pushes the SQL to the database immediately instead of waiting for the transaction to commit. Useful when you need a generated ID within the same transaction.

Q8. `findOne` / `getReferenceById` vs `findById`

`findById` hits the database and returns `Optional`. `getReferenceById` returns a **lazy proxy** without querying — useful when you only need to set a foreign key relationship and never read the object's fields. Touching a field on a proxy for a row that does not exist throws `EntityNotFoundException`.

✓ Check yourself before moving on

1. Explain the N+1 problem, how you detect it, and **four** ways to fix it.
2. Tell your 15s → 6s story in under 60 seconds, with the mechanism.
3. Explain why `@Transactional` does nothing on an internal method call.
4. State the default rollback behaviour and how to change it.
5. Give the fetch-type defaults for all four relationship annotations.

05 · Microservices

Time needed: 90 minutes

The JD lists "**Technology → Microservices → Microservices**" as a primary skill, and repeats it under preferred skills with **API Management**. Along with Spring Boot this is the most important technical file for this interview.

You have a real microservices project — the Classifieds platform with three services, Eureka and an API Gateway. Anchor your answers to it.

| First, the context: monolith vs microservices

The monolith

One deployable application containing everything: users, transactions, reporting, notifications. One codebase, one build, one database.

It is not a bad thing. It is simple to develop, simple to deploy, and a single transaction can span the whole system. Most applications should start here.

Where it hurts as it grows:

- One small change means redeploying everything.
- One memory leak in reporting takes down payments too.
- You cannot scale only the part that is busy — you scale the whole thing.
- A large team constantly blocks itself in one codebase.

Microservices

Split it into independent services, each owning one business capability, each with its own database, communicating over the network.

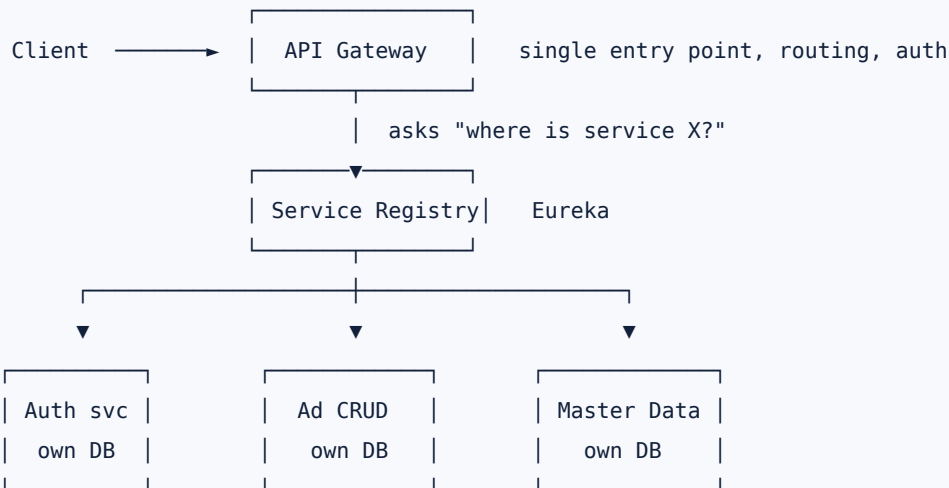
What you gain: independent deployment, independent scaling, fault isolation, and teams that can work without waiting for each other.

What you pay — say this part, because it shows judgement:

"You are trading code complexity for **operational and distributed-systems** complexity. Network calls fail, so you need retries and circuit breakers. There is no single transaction across services, so you need eventual consistency. Debugging spans several services, so you need distributed tracing and centralised logging. It is the right trade at a certain scale, but for a small application a well-structured monolith is usually the better engineering decision."

That last sentence is what separates a candidate who has *used* microservices from one who has *read* about them. Interviewers at Infosys hear the buzzword version all day.

Q1. • The core patterns — and which you used



1. Service Registry / Discovery — Eureka.

Services are on dynamic hosts and ports, and instances come and go. Hard-coding addresses does not work. Each service **registers** itself with Eureka on startup and sends heartbeats; callers **look up** the current instances by service name.

"Without discovery I would be hard-coding hostnames, which breaks the moment a service scales or moves. With Eureka a caller asks for **ad-service** and gets a live list."

2. API Gateway.

One entry point for all clients. It handles **routing**, and centralises the things you do not want to repeat in every service: authentication, rate limiting, CORS, request logging.

"Without a gateway the client would need to know every service's address, and every service would need its own copy of the auth logic. The gateway puts the cross-cutting concerns in one place."

(This is exactly the "Microservices API Management" the JD asks for. Say the words.)

3. Database per service.

Each service owns its data; no other service touches its tables. This is what makes independent deployment real — you can change your schema without coordinating with other teams.

The cost: no joins across services, and no single ACID transaction. Which leads to the next two patterns.

4. Config Server — externalised configuration for all services, so a change does not require a rebuild.

5. Circuit breaker — see Q3.

Q2. How do services talk to each other? ●

Synchronous — REST or gRPC. The caller waits for a reply.

```
@FeignClient(name = "master-data-service")    // declarative HTTP client
public interface MasterDataClient {
    @GetMapping("/categories/{id}")
    CategoryDto getCategory(@PathVariable Long id);
}
```

Simple, and you get an immediate answer. But it creates **temporal coupling**: if the callee is down, the caller fails too. Chains of synchronous calls multiply latency and failure probability.

Asynchronous — messaging over Kafka or RabbitMQ. The caller publishes an event and moves on.

"The publisher does not know or care who consumes it, and if a consumer is down the message waits in the queue. That removes the temporal coupling. The cost is eventual consistency — the effect is not immediate — and harder debugging, because the flow is no longer a straight line."

How to choose — the rule to state:

"If I need the answer to continue, synchronous. If I am telling the rest of the system that something happened, asynchronous. On the recall platform, publishing transaction state changes to Kafka for compliance logging and reporting was clearly the second kind — those consumers do not need to block the transaction from completing."

That is your project. Use it as the example.

Q3. ● Resilience — circuit breaker, retry, timeout

The problem: service A calls B, B becomes slow. A's threads pile up waiting. A runs out of threads and dies, so C which calls A dies too. One slow service takes down the system. That is a **cascading failure**.

Circuit breaker — three states, know all three:

State	Behaviour
Closed	Normal. Calls pass through, failures are counted.
Open	Failure threshold exceeded. Calls fail immediately without trying.
Half-open	After a wait, let a few test calls through. Success closes it, failure opens it again.

```
@CircuitBreaker(name = "masterData", fallbackMethod = "defaultCategory")
public CategoryDto getCategory(Long id) { return client.getCategory(id); }

public CategoryDto defaultCategory(Long id, Throwable t) {
    return CategoryDto.unknown();    // degrade gracefully instead of failing
}
```

Resilience4j is the current library; Hystrix is the older one, now retired.

"Failing fast protects both sides. The caller does not exhaust its threads waiting, and the struggling service gets breathing room to recover instead of being hammered."

Real-time example. The advertisement service calls the master data service for category names. If master data goes down without a circuit breaker, every ad request hangs on that call until it times out, the ad service's thread pool fills, and the ad service stops serving requests it could have answered perfectly well. With a breaker and a fallback, ads still render — just with a placeholder category. **Degraded is much better than down**, and that sentence is the point of the whole pattern.

Also mention **timeouts** (never make a network call without one) and **retry with exponential backoff and jitter** — and that you only retry **idempotent** operations, because retrying a payment could charge someone twice.

Q4. • Distributed transactions and the Saga pattern

The problem: placing an order must reserve stock, take payment and create a shipment — in three different services with three databases. There is no `@Transactional` that spans them.

Saga — break it into a sequence of local transactions, each publishing an event that triggers the next. If a step fails, run **compensating transactions** to undo the earlier ones.

```
Order created → Payment taken → Stock reserved → Shipment booked
                                x fails
                    ← refund payment ← cancel order          (compensation)
```

Two flavours: **choreography** (services react to each other's events — simple, but the overall flow is not written down anywhere) and **orchestration** (a coordinator drives the steps — clearer and easier to debug, but it is another component to run).

Mention **eventual consistency**: the system is briefly inconsistent, and the business has to accept that. In banking that decision belongs to the business, not the developer.

Real-time example. A recall spanning services would be: mark the transaction as recalled, reverse the ledger entry, notify the customer. If the ledger reversal fails after the transaction was already marked, you cannot simply roll back across two databases — you publish a compensating event that returns the transaction to its previous

state and records why. The compensation is itself an auditable business action, not a silent undo. In finance that is exactly how it has to work.

Idempotency matters here: a consumer may receive the same message twice, so processing must be safe to repeat. The usual approach is a unique message or request ID that you record and check before acting.

| Q5. Observability — the question people forget

"Once a request crosses five services, a stack trace is not enough. You need three things. **Centralised logging**, so all logs land in one place — ELK or Splunk. **Distributed tracing** with a correlation ID passed through every call, so you can follow one request end to end — Sleuth with Zipkin, or OpenTelemetry. And **metrics and health checks**, through Actuator and Prometheus, so you see latency and error rates per service."

Your resume mentions resolving production incidents by analysing logs, so this connects.

| Q6. Other things they may ask

How do you version an API? URL versioning (`/api/v1/...`) is the most common and the most visible. The real rule is to make **additive** changes — adding an optional field is backwards compatible; removing or renaming one is not.

How do you handle security between services? The gateway validates the JWT once, then services trust the network boundary, or pass the token along and each validates it. Service-to-service can use mTLS.

What is the Strangler Fig pattern? How you migrate a monolith gradually: put a proxy in front, move one capability at a time to a new service, and route that path to the new one. The monolith shrinks until it disappears. Nobody sensible does a big-bang rewrite.

What is a bounded context? From domain-driven design — the boundary within which a model has one consistent meaning. It is the sane way to decide **where to split services**: by business capability, not by technical layer.

"The wrong split is a 'database service' and a 'business logic service', because every change touches both. The right split is by business capability — orders, payments, notifications — so most changes stay inside one service."

How small should a service be? Not about lines of code:

"Small enough that one team can own it and understand it fully, and large enough that a typical change does not require modifying three services at once. If every feature spans multiple services, the boundaries are wrong."

Q7. Your Classifieds project — how to present it

Have this ready as a 90-second walkthrough:

"I built a three-service classifieds platform. An **authentication service** issuing and validating JWTs, an **advertisement service** with the CRUD operations, and a **master data service** for reference data like categories and locations. Each has its own database, so they can be deployed independently.

Eureka handles service discovery, so services resolve each other by name rather than hard-coded addresses. An **API Gateway** sits in front as the single entry point and does the routing, which also means authentication is handled in one place rather than duplicated in each service.

I added **global exception handling** with `@RestControllerAdvice` so all three services return the same error shape, and **DTO validation** with Bean Validation, so bad requests are rejected at the boundary rather than deep in the business logic. Security is a JWT layer with role-based access control across the services."

Be ready for the honest follow-ups:

- *"Did you deploy it?"* — answer truthfully. If it ran locally with Docker Compose, say that.
- *"How did you handle a service being down?"* — if you did not add a circuit breaker, say so and describe what you would add now. That is a better answer than inventing one.
- *"Why three services and not one?"* — it is a learning project; say so, and then show you know the trade-offs from the section above. Honesty plus understanding beats a bluff that collapses.

✓ Check yourself before moving on

1. Explain monolith vs microservices **including what you give up**.
2. Explain the three circuit breaker states and why failing fast helps both sides.
3. Explain the Saga pattern and compensating transactions.
4. Explain why you would use messaging instead of a REST call, with your Kafka example.
5. Give the 90-second Classifieds walkthrough without notes.

06 · Kafka and RabbitMQ

Time needed: 60 minutes

This is your **differentiator**. Most candidates at this level have Spring Boot and Hibernate. Far fewer have used both Kafka and RabbitMQ in production, and your resume says you used each for a *different reason*. That is the interesting part — be ready to explain the reasoning, not just name the tools.

| First, the context: why messaging at all?

The problem you actually solved

An approval endpoint that sends an email synchronously:

```
@PostMapping("/approve/{id}")
public ResponseEntity<?> approve(@PathVariable Long id) {
    cardService.approve(id);           // 200 ms
    emailService.send(...);           // 2-4 seconds, and it can fail or hang
    return ResponseEntity.ok();       // the user has been waiting the whole time
}
```

Three problems:

1. **The user waits** for something they do not care about.
2. **If the mail server is slow, the API times out** — and to the user it looks like the approval failed, even though it succeeded.
3. **Bulk approvals multiply it**. 50 approvals means 50 sequential email sends inside the request.

The fix

```
@PostMapping("/approve/{id}")
public ResponseEntity<?> approve(@PathVariable Long id) {
    cardService.approve(id);           // 200 ms
    rabbitTemplate.convertAndSend("email.queue", new EmailMessage(...)); // ~1 ms
    return ResponseEntity.ok();       // returns immediately
}
```

A separate consumer picks the message up and sends the email. The API responds in milliseconds. If the mail server is down, the message waits in the queue and is delivered when it recovers, instead of being lost.

Say it this way:

"The email is not part of what the user is waiting for, so it should not be inside the request. Moving it to RabbitMQ meant the API returned immediately, the timeouts during bulk approvals stopped, and a mail server outage no longer failed the approval — the messages just queued up until it came back."

That is exactly what your resume says you did. Make sure the *reasoning* comes out, not just the tool name.

Q1. ●● Kafka vs RabbitMQ — when do you use which?

This is the question your resume invites. You used both, so you will be asked to justify it.

	RabbitMQ	Kafka
Model	Message broker / queue	Distributed event log
After consumption	Message is removed	Message stays for its retention period
Consumers	Usually one consumer per message	Many independent consumers read the same events
Replay	No	Yes — reset the offset and re-read
Routing	Rich — exchanges, routing keys, topics	Simple — topics and partitions
Ordering	Per queue	Per partition
Throughput	High	Very high — designed for streams
Best for	Task queues, work distribution, RPC-style	Event streaming, audit trails, multiple subscribers

The one-line distinction to memorise:

"RabbitMQ is a **postbox** — a message is delivered to a worker and then it is gone. Kafka is a **ledger** — events are appended and stay there, and any number of consumers can read them at their own pace, including re-reading from the past."

● Your answer, using your own project

This is the strongest version, because it is true and it shows you chose deliberately:

"I used them for two different jobs.

RabbitMQ for the email notifications. That is a task queue: one message, one worker, do the job, and the message is done. I did not need to keep it or replay it — I needed the work off the request thread so bulk approvals stopped timing out.

Kafka for publishing transaction state changes. That is an event stream, and several downstream consumers needed it — compliance logging and report generation, and potentially more later. With Kafka each consumer group reads independently at its own pace, and because the events are retained, a consumer that was down can catch up, and compliance can replay history if it needs to rebuild. With a queue, the first consumer would take the message and the others would never see it."

That last sentence is the technical heart of it. It shows you understand the difference rather than having used whatever the team had.

Q2. RabbitMQ concepts

Producer → Exchange → (routing rules) → Queue → Consumer

The **exchange** is the piece people forget. A producer never sends to a queue directly; it sends to an exchange, which decides which queues get a copy.

Exchange type	Routes by
Direct	Exact routing key match
Topic	Wildcard pattern — <code>txn.*.approved</code>
Fanout	Copies to every bound queue, ignoring the key
Headers	Header values

```
@RabbitListener(queues = "email.queue")
public void handle(EmailMessage msg) {
    emailSender.send(msg);           // ack on normal return; requeue or DLQ on exception
}
```

Acknowledgements. The broker only removes a message once the consumer acknowledges it. If the consumer crashes mid-processing, the message is redelivered — so processing must be **idempotent**.

Dead Letter Queue (DLQ). Where messages go after repeated failures, instead of being retried forever or dropped silently. Mentioning DLQs unprompted signals production experience:

"Without a DLQ a poison message either blocks the queue or gets retried indefinitely. With one, it moves aside after N attempts, gets alerted on, and someone can inspect it without holding up everything behind it."

Q3. Kafka concepts ●

Topic: transaction-events

```
└─ Partition 0 → [ msg ][ msg ][ msg ]      ← ordered within the partition
└─ Partition 1 → [ msg ][ msg ]
└─ Partition 2 → [ msg ][ msg ][ msg ]
```

Topic — a named stream of events.

Partition — a topic is split into partitions, which is how Kafka scales. Order is guaranteed **within** a partition, not across the topic.

Offset — each consumer's position in a partition. Consumers commit their offsets, so they can resume where they left off.

Consumer group — consumers in the same group **share** the partitions, one partition per consumer at a time. Consumers in **different** groups each get **all** the messages.

That last point is the mechanism behind your answer in Q1: compliance and reporting are two different consumer groups, so both see every event.

● How do you guarantee ordering in Kafka?

"By partition key. All messages with the same key go to the same partition, and order is guaranteed within a partition. So for transaction events I would key by account or transaction ID, which guarantees that everything about one transaction is processed in order, while different transactions still spread across partitions for throughput."

Delivery guarantees — know the three:

- **At most once** — commit the offset before processing. Fast, may lose messages.
- **At least once** — commit after processing. **The common default.** May duplicate, so consumers must be idempotent.
- **Exactly once** — possible with Kafka transactions, but costly and complex.

"In practice most systems run at-least-once and make consumers idempotent, because exactly-once across a distributed system is expensive. For financial events I would record a processed-message ID and check it before acting, so a redelivery is a no-op."

Replication. Each partition has a leader and followers on other brokers. If the leader dies, a follower is promoted. That is Kafka's durability story.

Q4. Quick comparison questions

"Why not just use a database table as a queue?"

"It works at small scale, and people do it. But you end up building polling, locking, retries, dead-lettering and back-pressure yourself, and polling a table is wasteful and adds latency. A broker gives you all of that, plus push delivery."

"What happens if a message fails?"

Retry with backoff, then a dead letter queue after N attempts, then alert. Never silently swallow.

"What if the same message arrives twice?"

Idempotency: a unique message ID recorded on the consumer side, or design the operation so applying it twice has the same effect as once.

"How do you not lose messages?"

Producer side: acknowledgements and retries. Broker side: persistence and replication. Consumer side: acknowledge only **after** successful processing.

✓ Check yourself before moving on

1. Kafka vs RabbitMQ in one sentence each, then **your** reason for using both.
2. Explain consumer groups and why two groups both receive every event.
3. Explain how ordering works in Kafka and how you would key transaction events.
4. Explain why at-least-once delivery means consumers must be idempotent.
5. Tell the email-timeout story with the reasoning, in 45 seconds.

07 · Spring Security and JWT

Time needed: 45 minutes

Your resume lists Spring Security with role-based access control, and JWT in your microservices project. Expect a few questions, especially in a banking context.

| First, the context: authentication vs authorisation

- **Authentication** — *who are you?* Checking credentials.
- **Authorisation** — *what are you allowed to do?* Checking permissions.

You authenticate once; you authorise on every request. In your Business Card system, authentication proved the user was a bank employee; authorisation decided whether that employee was a **requester**, an **approver** or an **admin**.

| Q1. How does Spring Security work?

It sits in front of your application as a **chain of servlet filters**. Every request passes through the chain before reaching a controller.

```
Request → [ filters: authenticate, authorise, CSRF, CORS ] → DispatcherServlet → Controller
```

The key filter for a token-based API is one that reads the token, validates it, and places an `Authentication` object into the `SecurityContext`. Everything downstream then knows who the caller is.


```

@Configuration
@EnableWebSecurity
public class SecurityConfig {

    @Bean
    SecurityFilterChain chain(HttpSecurity http) throws Exception {
        http
            .csrf(csrf -> csrf.disable()) // stateless API - see Q5
            .sessionManagement(s -> s.sessionCreationPolicy(SessionCreationPolicy.STATELESS))
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/api/auth/**").permitAll()
                .requestMatchers("/api/cards/approve/**").hasRole("APPROVER")
                .requestMatchers("/api/admin/**").hasRole("ADMIN")
                .anyRequest().authenticated()
            )
            .addFilterBefore(jwtFilter, UsernamePasswordAuthenticationFilter.class);
        return http.build();
    }
}

```

Q2. • Role-based access control — your story

Two ways to enforce it. Know both.

At the URL level, as above with `hasRole(...)`.

At the method level, which is more precise:

```

@PreAuthorize("hasRole('APPROVER')")
public CardDto approve(Long id) { ... }

@PreAuthorize("hasRole('ADMIN') or #userId == authentication.principal.id")
public UserDto getUser(Long userId) { ... } // admins, or the user themselves

```

Requires `@EnableMethodSecurity`.

Your version:

"In the Business Card system there were three roles — requester, approver and admin — and the rule that mattered was that a requester must never be able to approve. I enforced it at the API layer with Spring Security rather than in the UI, because a UI check only hides the button. Anyone can call the endpoint directly with a REST client, so the authorisation has to be server side."

That last sentence is the point. In a banking interview it lands well.

⚠ **hasRole vs hasAuthority** : `hasRole("ADMIN")` looks for the authority `ROLE_ADMIN` — Spring adds the prefix for you. `hasAuthority("ROLE_ADMIN")` is the literal form. Mismatched prefixes are a very common bug.

Q3. • JWT — how it works

A JWT is three base64 sections separated by dots:

```
header.payload.signature
```

```
header    { "alg": "HS256", "typ": "JWT" }  
payload   { "sub": "neelima", "roles": ["APPROVER"], "exp": 1735689600 }  
signature HMACSHA256(base64(header) + "." + base64(payload), secret)
```

The flow

1. POST /login with credentials
2. Server verifies, signs a JWT, **returns** it
3. Client sends it **on** every request: Authorization: Bearer <token>
4. Server verifies the SIGNATURE and expiry - no **data**base lookup needed
5. Request proceeds with the user's identity and roles

• Why use a JWT instead of a session?

"A session means the server stores state and every request looks it up, which is fine for one server but awkward when you scale — you need sticky sessions or a shared store like Redis. A JWT is **self-contained**: the user and roles are inside it, and the signature proves it has not been tampered with. So any instance of any service can validate it independently, with no shared session store. In a microservices setup that is the deciding factor."

• The two things people get wrong

1. A JWT is signed, not encrypted. Anyone can base64-decode the payload and read it. The signature only proves it has not been *altered*. **Never put anything sensitive in the payload** — no passwords, no account numbers.

2. You cannot easily revoke one. It is valid until it expires, so if a token is stolen it works until then. The usual answers: keep access tokens **short-lived** (15 minutes) with a longer-lived **refresh token**, and keep a small **blocklist** of revoked token IDs for logout. Say the trade-off rather than pretending it is solved.

Q4. Where should the token be stored on the client?

"An `httpOnly` cookie is safer than `localStorage`, because JavaScript cannot read an `httpOnly` cookie, so an injected script cannot steal it. Anything in `localStorage` is readable by any script on the page. The trade-off is that cookies need CSRF protection, whereas an `Authorization` header does not."

Q5. CSRF, and why it is disabled for APIs

CSRF is when another site makes the user's browser send a request to yours, relying on the browser automatically attaching the session cookie.

"It is disabled for a stateless API because the attack depends on the browser attaching credentials automatically. If the client sends an explicit `Authorization` header, another site cannot make the browser add it. If I were using cookie-based auth, CSRF protection would need to stay on."

Do not just say "we disable CSRF" — say *why*, or it looks like copied configuration.

Q6. Password storage

```
@Bean PasswordEncoder passwordEncoder() { return new BCryptPasswordEncoder(); }
```

Never store plain text, and never a plain hash like MD5 or SHA-256. BCrypt is deliberately **slow** and **salts** each password automatically, which defeats rainbow tables and makes brute force expensive.

Q7. Other things worth a sentence

- **OAuth2 vs JWT** — not alternatives. OAuth2 is the authorisation *framework*; a JWT is a token *format* often used within it.
- **CORS vs CSRF** — CORS is a browser rule about which origins may read a response. CSRF is an attack. They get confused constantly.
- **Method security vs URL security** — URL rules are coarse and can be bypassed if another entry point reaches the same service method. Method-level annotations protect the method itself.

✓ Check yourself before moving on

1. Explain the JWT structure and the full request flow.
2. Explain why a JWT suits microservices better than server-side sessions.
3. Explain why a JWT is not secret, and what that means for what you put in it.
4. Explain why CSRF is disabled for a stateless API.
5. Tell your RBAC story, ending on why authorisation must be server side.

08 · SQL and Query Optimisation

Time needed: 75 minutes

Your resume claims SQL tuning, indexing and execution plan analysis. That is unusual at this level and it is genuinely valuable — but it also means you **will** be tested on it. An interviewer who sees "identified and removed N+1 queries, added strategic indexing" will ask you to explain indexing properly.

Q1. Joins — know all four

```
-- INNER: only rows that match in both tables
SELECT t.id, c.name FROM transactions t
INNER JOIN customers c ON t.customer_id = c.id;

-- LEFT: every row from the left, NULLs where the right has no match
SELECT c.name, t.id FROM customers c
LEFT JOIN transactions t ON t.customer_id = c.id;    -- includes customers with no txns

-- RIGHT: the mirror image. Rarely used - people rewrite it as a LEFT JOIN.
-- FULL OUTER: everything from both sides. MySQL does not support it directly.
```

The classic question: "Find customers with no transactions."

```
SELECT c.* FROM customers c
LEFT JOIN transactions t ON t.customer_id = c.id
WHERE t.id IS NULL;
```

The `WHERE ... IS NULL` after a LEFT JOIN is the pattern. Learn it.

⚠ **A trap worth knowing:** putting a condition on the right-hand table in the `WHERE` clause of a LEFT JOIN silently turns it into an INNER JOIN, because `NULL` fails the condition. If you want to filter the joined table without losing left rows, the condition belongs in the `ON` clause.

Q2. ●● Indexing — the answer that matters most here

What an index actually is

"An index is a separate sorted structure, usually a **B-tree**, that maps column values to row locations. Without one, the database does a **full table scan** — it reads every row to find matches, which is $O(n)$. With one, it walks the tree, which is $O(\log n)$. It is the same reason you use the index at the back of a book instead of reading every page."

When an index helps

- Columns in a `WHERE` clause
- Columns you `JOIN` on — **especially foreign keys, which are not indexed automatically in MySQL's InnoDB unless you add them**
- Columns in `ORDER BY` and `GROUP BY`

• The cost — always mention this, it is what shows real understanding

"Indexes are not free. Every `INSERT`, `UPDATE` and `DELETE` has to update every index on that table, so over-indexing slows down writes. They also take disk space. On a write-heavy table I index deliberately, not by default — the right number is the smallest set that serves the actual query patterns."

Saying only "indexes make queries faster" is the answer of someone who read it. Saying "and they slow down writes, so I add them deliberately" is the answer of someone who has tuned a real system.

Composite indexes and the left-most prefix rule

```
CREATE INDEX idx_txn_status_date ON transactions (status, created_at);
```

This index serves:

- `WHERE status = 'PENDING'` ✓
- `WHERE status = 'PENDING' AND created_at > '2026-01-01'` ✓

But **not** `WHERE created_at > '2026-01-01'` alone ✗ — because the index is sorted by `status` first. This is the **left-most prefix rule**, and it is a favourite question. Column order in a composite index matters.

Real-time example. The recall report always filters by status and then by a date range: `WHERE status = 'PENDING' AND created_at BETWEEN ? AND ?`. So the index is `(status, created_at)` in that order — the equality column first, the range column second. Reversed, as `(created_at, status)`, the index is far less effective, because the range scan happens first and status cannot narrow it. Getting that column order right was part of the 15-second to 6-second fix.

What stops an index being used

```
WHERE YEAR(created_at) = 2026      -- ✗ a function on the column defeats the index
WHERE created_at >= '2026-01-01'  -- ✓ same meaning, index usable

WHERE account_number LIKE '%1234'  -- ✗ a leading wildcard cannot use a B-tree
WHERE account_number LIKE '1234%'  -- ✓ prefix match is fine
```

Mentioning these two unprompted is a strong signal.

Clustered vs non-clustered

- **Clustered** — the table rows are physically stored in this order. One per table; in InnoDB it is the primary key.
- **Non-clustered** — a separate structure pointing back to the row.

Covering index

An index containing every column a query needs, so the database answers it from the index alone and never touches the table. Sometimes the biggest single win available.

Real-time example. A dashboard count of pending recalls per branch needs only `status` and `branch_code`. An index on `(status, branch_code)` covers it, so the database answers from the index and never reads a single table row. On a large transactions table that is the difference between milliseconds and seconds.

Q3. • EXPLAIN — how you actually diagnose

Your resume says you analysed execution plans. Be ready to describe the process.

```
EXPLAIN SELECT t.id, c.name FROM transactions t
JOIN customers c ON t.customer_id = c.id
WHERE t.status = 'PENDING';
```

What you look for:

Column	What matters
type	<code>ALL</code> = full table scan, the thing to fix. <code>ref</code> or <code>range</code> is good. <code>const</code> / <code>eq_ref</code> is best.
key	Which index was actually used. <code>NULL</code> means none.
rows	Estimated rows examined. A big number here is the problem.
Extra	<code>Using filesort</code> or <code>Using temporary</code> mean extra work. <code>Using index</code> is a covering index — good.

Say the process, not just the tool:

"I start from the slow query log or the application timings to find *which* query is slow — there is no point optimising the wrong one. Then I run `EXPLAIN` on it. If the type is `ALL` and the row count is large, it is doing a full scan, and I look at what the `WHERE` and `JOIN` columns are and whether an index exists. After adding one I re-run `EXPLAIN` to confirm it is actually being used, and re-measure the query time. Adding an index without checking the plan afterwards is guessing."

Q4. Normalisation, and when to break it

- **1NF** — atomic values, no repeating groups.
- **2NF** — 1NF, and no partial dependency on part of a composite key.

- **3NF** — 2NF, and no non-key column depending on another non-key column.

"Normalisation removes duplication, so there is one place to update a fact and no inconsistency. The cost is more joins. **Denormalisation** deliberately duplicates data to avoid expensive joins on read-heavy paths — for example storing a customer name on a report table so a reporting query does not join. It is a trade: faster reads, but now you have two copies to keep in sync."

Q5. Common query questions

Second-highest salary — asked constantly:

```
SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);

-- or, more generally, the Nth highest
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 1;
```

Find duplicates:

```
SELECT email, COUNT(*) FROM customers GROUP BY email HAVING COUNT(*) > 1;
```

• **WHERE vs HAVING** : **WHERE** filters **rows before** grouping. **HAVING** filters **groups after** aggregation. You cannot use an aggregate in **WHERE**.

UNION vs UNION ALL : **UNION** removes duplicates, which costs a sort. **UNION ALL** keeps everything and is faster. Use **UNION ALL** unless you actually need deduplication.

DELETE vs TRUNCATE vs DROP : **DELETE** removes rows, can have a **WHERE**, is logged and can be rolled back. **TRUNCATE** empties the table quickly, no **WHERE**, resets auto-increment. **DROP** removes the table itself.

Window functions, worth knowing one:

```
SELECT name, salary, department,
       RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rank_in_dept
FROM employees;
```

Unlike **GROUP BY**, a window function keeps every row and adds a computed column.

Q6. ACID

- **Atomicity** — all of the transaction, or none of it.
- **Consistency** — the database moves from one valid state to another; constraints hold.
- **Isolation** — concurrent transactions do not corrupt each other.
- **Durability** — once committed, it survives a crash.

The banking example makes this concrete: a transfer is a debit and a credit. Without atomicity, a crash between them destroys money. Use that example — it fits your domain.

Q7. SQL vs NoSQL — you have MongoDB on your resume

"I would use a relational database when the data is structured and relational and I need ACID transactions — which is most of banking. MongoDB suits flexible or evolving schemas, document-shaped data, and cases where horizontal scaling matters more than joins. The real question is the access pattern: if I am constantly joining, that is a relational shape, and forcing it into documents means duplicating data everywhere."

Q8. Pagination at the database level

```
SELECT * FROM transactions ORDER BY created_at DESC LIMIT 20 OFFSET 0;
```

"OFFSET pagination degrades on deep pages, because the database still has to scan and discard everything before the offset. Page 5,000 is slow. **Keyset pagination** — `WHERE created_at < :lastSeen ORDER BY created_at DESC LIMIT 20` — stays fast because the index seeks straight to the position."

That is a strong detail for a reporting-heavy system, which yours was.

✓ Check yourself before moving on

1. Explain what an index is, **and its cost on writes**.
2. Explain the left-most prefix rule with the composite index example.
3. Give two things that stop an index being used.
4. Describe your EXPLAIN process end to end.
5. Write the "customers with no transactions" query from memory.
6. Explain `WHERE` vs `HAVING`.

09 · Coding Round

Time needed: 90 minutes. Write the code, do not read it.

Reading solutions feels productive and is nearly worthless. Open an IDE and type these from an empty file.

| What to expect at Infosys

The coding bar for an experienced hire at a services company is **moderate**. They are not asking for dynamic programming. Expect:

- String and array manipulation
- Collections and HashMap usage
- Java 8 stream problems
- One "write a small class" or design question
- Sometimes SQL instead of code

They are checking: can you write correct, compiling Java; do you handle edge cases; do you talk while you work. Not whether you know a clever trick.

The process, which is half the marks:

1. **Restate the problem.** "So I need to return the first character that appears once, and null if there isn't one?"
 2. **Ask about edge cases** *before* coding. Empty input? Nulls? Case sensitivity?
 3. **Say your approach out loud.** "I'll count with a LinkedHashMap so I keep order, then find the first with a count of one."
 4. **Then write it.**
 5. **Trace one example** out loud when done.
 6. **State the complexity.** They almost always ask, so beat them to it.
-

| Problem 1 ● — First non-repeating character

The most common Java string question in Indian interviews.

```

public static Character firstNonRepeating(String input) {
    if (input == null || input.isEmpty()) return null;

    Map<Character, Integer> counts = new LinkedHashMap<>(); // preserves insertion order
    for (char c : input.toCharArray()) {
        counts.merge(c, 1, Integer::sum);
    }

    for (Map.Entry<Character, Integer> e : counts.entrySet()) {
        if (e.getValue() == 1) return e.getKey();
    }

    return null;
}

```

Say: "LinkedHashMap, not HashMap, because I need insertion order to find the *first* one. Two passes, so $O(n)$ time and $O(k)$ space where k is the distinct character count."

Problem 2 ● — Reverse a string, and check a palindrome

```

public static String reverse(String s) {
    if (s == null) return null;
    return new StringBuilder(s).reverse().toString();
}

// they often want it without the library method
public static String reverseManual(String s) {
    char[] chars = s.toCharArray();
    int left = 0, right = chars.length - 1;
    while (left < right) {
        char tmp = chars[left];
        chars[left++] = chars[right];
        chars[right--] = tmp;
    }
    return new String(chars);
}

public static boolean isPalindrome(String s) {
    String clean = s.replaceAll("[^A-Za-z0-9]", "").toLowerCase();
    int left = 0, right = clean.length() - 1;
    while (left < right) {
        if (clean.charAt(left++) != clean.charAt(right--)) return false;
    }
    return true;
}

```

The two-pointer palindrome is better than reversing and comparing, because it can exit early and uses no extra string. Say that.

Problem 3 ● — Duplicates in an array

```
// find duplicates
public static Set<Integer> findDuplicates(int[] nums) {
    Set<Integer> seen = new HashSet<>();
    Set<Integer> dupes = new HashSet<>();
    for (int n : nums) {
        if (!seen.add(n)) dupes.add(n);    // add() returns false if already present
    }
    return dupes;
}

// remove duplicates, keeping order
public static List<Integer> removeDuplicates(List<Integer> nums) {
    return new ArrayList<>(new LinkedHashSet<>(nums));
}
```

`seen.add(n)` returning false is the neat part — one lookup instead of `contains` then `add`.

Problem 4 ● — Stream problems

These are very likely, because your resume says Java 8.

```

// count word frequency
Map<String, Long> freq = Arrays.stream(sentence.toLowerCase().split("\\s+"))
    .collect(Collectors.groupingBy(w -> w, Collectors.counting()));

// second highest number
Optional<Integer> second = nums.stream()
    .distinct()
    .sorted(Comparator.reverseOrder())
    .skip(1)
    .findFirst();

// group employees by department
Map<String, List<Employee>> byDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment));

// highest paid employee per department
Map<String, Optional<Employee>> topPerDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment,
        Collectors.maxBy(Comparator.comparing(Employee::getSalary))));

// sum of even numbers
int sum = nums.stream().filter(n -> n % 2 == 0).mapToInt(Integer::intValue).sum();

// sort a map by value, descending
Map<String, Integer> sorted = map.entrySet().stream()
    .sorted(Map.Entry.<String, Integer>comparingByValue().reversed())
    .collect(Collectors.toMap(Map.Entry::getKey, Map.Entry::getValue,
        (a, b) -> a, LinkedHashMap::new));

```

That last one is worth practising — the four-argument `toMap` with `LinkedHashMap::new` is needed to keep the sort order, and people always forget it.

Problem 5 — Two sum

```

public static int[] twoSum(int[] nums, int target) {
    Map<Integer, Integer> seen = new HashMap<>();    // value -> index
    for (int i = 0; i < nums.length; i++) {
        int need = target - nums[i];
        if (seen.containsKey(need)) return new int[]{ seen.get(need), i };
        seen.put(nums[i], i);
    }
    return new int[]{};
}

```

Say: "The brute force is two nested loops, $O(n^2)$. Storing what I have seen in a HashMap makes the lookup $O(1)$, so it becomes one pass, $O(n)$ time and $O(n)$ space. That is the classic time-for-space trade."

Talking about the brute force *first*, then improving it, is exactly what they want.

Problem 6 — Fibonacci and factorial

```
// iterative - preferred, no stack growth
public static long fib(int n) {
    if (n <= 1) return n;
    long prev = 0, curr = 1;
    for (int i = 2; i <= n; i++) {
        long next = prev + curr;
        prev = curr;
        curr = next;
    }
    return curr;
}

// recursive with memoisation, if they ask for recursion
public static long fibMemo(int n, Map<Integer, Long> memo) {
    if (n <= 1) return n;
    return memo.computeIfAbsent(n, k -> fibMemo(k - 1, memo) + fibMemo(k - 2, memo));
}
```

Naive recursion is $O(2^n)$ and will be called out. Memoisation makes it $O(n)$.

Problem 7 — Reverse words, count vowels, anagram

```
public static String reverseWords(String s) {
    String[] words = s.trim().split("\\s+");
    Collections.reverse(Arrays.asList(words));
    return String.join(" ", words);
}

public static boolean isAnagram(String a, String b) {
    if (a == null || b == null || a.length() != b.length()) return false;
    char[] x = a.toCharArray(), y = b.toCharArray();
    Arrays.sort(x); Arrays.sort(y);
    return Arrays.equals(x, y);
}
```

For anagram, mention the alternative: counting characters in a map is $O(n)$ versus $O(n \log n)$ for sorting. Naming the trade-off scores.

Problem 8 — "Write a class" questions

Singleton, thread safe:

```

public class ConfigManager {
    private static volatile ConfigManager instance;
    private ConfigManager() { }

    public static ConfigManager getInstance() {
        if (instance == null) { // first check, no lock
            synchronized (ConfigManager.class) {
                if (instance == null) { // second check, with lock
                    instance = new ConfigManager();
                }
            }
        }
        return instance;
    }
}

```

This is **double-checked locking**. `volatile` is essential — without it another thread can see a partially constructed object due to instruction reordering.

Say the better answer too: "In a Spring application I would not write this at all. A `@Component` is already a singleton in the container, and that is easier to test than a static instance. The enum singleton is also simpler if I truly need one outside Spring."

Immutable class:

```

public final class Money { // final: cannot be subclassed
    private final BigDecimal amount; // final fields
    private final String currency;

    public Money(BigDecimal amount, String currency) {
        this.amount = amount;
        this.currency = currency;
    }
    public BigDecimal getAmount() { return amount; } // no setters
}

```

The four rules: make the class `final`, all fields `private final`, no setters, and if a field is mutable (a `Date`, a `List`) return a **defensive copy** from the getter rather than the field itself.

Java-specific traps they enjoy

"What does this print?"

```
Integer a = 127, b = 127;  
Integer c = 128, d = 128;  
System.out.println(a == b); // true - Integer cache covers -128..127  
System.out.println(c == d); // false - outside the cache, different objects  
System.out.println(c.equals(d)); // true
```

```
String s1 = "abc";  
String s2 = "abc";  
String s3 = new String("abc");  
s1 == s2 // true - string pool  
s1 == s3 // false  
s1.equals(s3) // true  
s1 == s3.intern() // true - intern() returns the pooled instance
```

```
System.out.println(0.1 + 0.2 == 0.3); // false - floating point  
// which is exactly why money uses BigDecimal
```

✓ Practice plan for this file

Type problems **1, 3, 4 and 5** from an empty file, no copy-paste. If you can write those four while explaining what you are doing, you can handle this round.

10 · Backend System Design

Time needed: 75 minutes

At this level Infosys will not ask you to design Netflix. But the JD is full of consulting language — *"problem definition, effort estimation, diagnosis, solution generation and design"* — so expect one design question, and expect them to be watching **how you think**, not whether you produce a perfect architecture.

The framework — use it for every design question

- | | |
|----------------------|---|
| 1. CLARIFY | Ask 2-3 questions. Never start designing immediately. |
| 2. SCOPE | State the core use cases you will design for . |
| 3. API | What end points, what request and response shapes. |
| 4. DATA MODEL | Tables or documents, and the key relationships. |
| 5. FLOW | Walk one request end to end . |
| 6. SCALE | What breaks first, and what you would do about it. |

Step 1 is the one candidates skip, and it is the one the JD explicitly asks for.

"Problem definition" is literally in the job description.

Clarifying questions that fit almost anything:

- "Roughly what volume are we designing for — hundreds a day or millions?"
- "Is this read-heavy or write-heavy?"
- "Does this need to be strongly consistent, or is eventual consistency acceptable?"
- "Is there an existing system I am integrating with, or is this greenfield?"

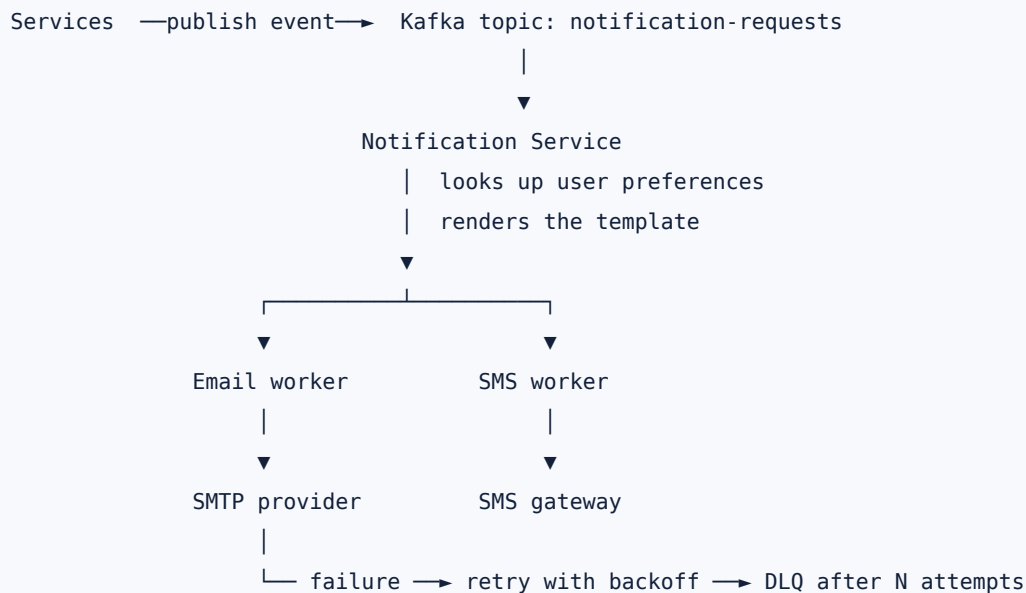
That third question is a strong one to ask in a banking context, because the answer is usually "strongly consistent for money, eventually consistent for notifications" — and saying that shows domain judgement.

Design 1 ● — A notification service

Most likely for you, because your resume already describes half of it.

Clarify: Which channels — email, SMS, push? Do we need delivery status? Are there priorities, or is everything equal? Roughly what volume?

The design:



The points to make:

- **Asynchronous, always.** The calling service must never wait for an email. This is exactly what you did with RabbitMQ.
- **Preferences and templates** live in the notification service, not scattered across callers. One place to change how something is worded.
- **Retry with exponential backoff**, then a dead letter queue, then an alert. Never silently drop a notification.
- **Idempotency**: a duplicate event must not send two emails. Key on an event ID and record what has been sent.
- **Delivery status** written back so support can answer "did the customer get it?"

Then land it: *"This is close to what I built — the async part with RabbitMQ for emails, and Kafka for publishing state changes to several consumers."*

Design 2 • — A transaction recall / approval workflow

Your actual domain. If they let you pick, pick this.

Clarify: Who can raise a recall, who approves? Is there a time limit? Does it need an audit trail? Single or multi-level approval?

Data model:

```

recall_request
  id, transaction_id, requested_by, reason, status, created_at, updated_at

recall_approval
  id, recall_request_id, approver_id, decision, comments, decided_at

audit_log
  id, entity_type, entity_id, action, actor, before_state, after_state, timestamp
  
```

The state machine — draw it, it always impresses:



The points to make:

- **Model the states explicitly** and validate transitions in one place. An approved request must never go back to draft. *(Your resume mentions unit tests for state-transition validation — say that here.)*
- **Authorisation server side.** A requester must not be able to approve their own request. Enforce in the service layer, not the UI.
- **Append-only audit log.** In banking every state change must be attributable — who, what, when, and what it was before. Never update audit rows.
- **Publish each state change as an event** for compliance and reporting consumers. That is your Kafka work.
- **Idempotency on submit**, so a double-click does not create two recall requests.

Design 3 — A URL shortener

The classic warm-up. Know it in case it comes up.

Core: `POST /shorten` takes a long URL, returns a short code. `GET /{code}` returns a **301 or 302 redirect**.

Generating the code: a base62 encoding of an auto-increment ID is simple and collision-free. Hashing the URL is an alternative but needs collision handling.

Scale points: it is extremely read-heavy, so cache the code-to-URL mapping in Redis in front of the database. The lookup must be a primary key or unique index hit. Mention 301 (permanent, browser caches it, so you lose click analytics) versus 302 (temporary, every click reaches you) — that trade-off is the interesting part.

Design 4 — "How would you design a REST API for X?"

The generic answer, which is worth having ready:

GET	/api/v1/transactions	list, with pagination and filters
GET	/api/v1/transactions/{id}	one
POST	/api/v1/transactions	create → 201 + Location header
PUT	/api/v1/transactions/{id}	full replace
PATCH	/api/v1/transactions/{id}	partial update
DELETE	/api/v1/transactions/{id}	remove → 204

The rules to state:

- **Nouns, not verbs.** `/transactions/{id}/recall` as a sub-resource, not `/recallTransaction?id=`.
- **Correct status codes.** 201 for created, 204 for no content, 400 for bad input, 401 vs 403, 404, 409 for a conflict, 422 for validation, 500 for server errors.

- **Pagination on every list endpoint.** Never return an unbounded collection.
- **Versioning** in the URL, and only additive changes within a version.
- **A consistent error shape** across every endpoint, via `@RestControllerAdvice`.
- **Idempotency** for non-safe operations, usually via a client-supplied key.
- **DTOs, never entities, at the boundary.** Returning entities leaks your schema, causes lazy-loading surprises during serialisation, and means a database change breaks your API contract.

That last point is a genuinely good one and few candidates say it.

Design 5 — "This API is slow. What do you do?"

You have lived this. Give a process, not a list of fixes.

1. MEASURE Where is the time actually **going**? Application logs and timings first, **then** the slow query log. Do not guess.
2. CATEGORISE Is it the **data**base, the application, or an external call?
 - **Database** → N+1, missing index, bad JOIN
 - Application → inefficient loop, unnecessary object creation
 - External → a synchronous call that should be async
3. FIX THE BIGGEST **ONE** FIRST
4. RE-MEASURE Prove it. Compare before and after.

Then make it concrete: *"That is exactly the process that took the report from 15 seconds to 6. The measurement step told me it was N+1 rather than the query itself, which I would not have guessed."*

The honest line that scores: *"My first assumption about the cause was wrong, which is why I measure before optimising."*

✓ Check yourself before moving on

1. Recite the six-step framework.
2. Design the notification service out loud in 3 minutes, including retries and DLQ.
3. Draw the recall state machine and say why the audit log is append-only.
4. Give five REST API design rules, including DTOs at the boundary.

11 · Talking About Your Experience

Time needed: 60 minutes, said out loud

| First, the context: your real position

You are stronger for this role than you probably feel, for three specific reasons.

1. You have real numbers. 15 seconds to 6 seconds. 2-3 seconds to under 1 second. Most candidates say "I optimised the API." You can say what was wrong, what you changed, and what it became. In a services company interview that is rare and it is remembered.

2. You have banking domain experience. The JD asks for "one or two industry domain knowledge". You worked on a UK private bank's transaction recall and card management systems. Infosys staffs projects by domain, and someone who already understands banking workflows is easier to place. Easier to place means easier to hire.

3. You used Kafka and RabbitMQ for different reasons. Not just "we had messaging" — a task queue for one job and an event log for another. That is a design decision, and it is the most senior-sounding thing on your resume.

What to watch: your experience is about 1 year 9 months of engineering. Do not oversell it (see section 5), and do not undersell it either. Depth on the things you actually did beats vague claims about things you did not.

| • "Tell me about yourself" — your script

Structure: **Now → What you have built → Why this role.** Under two minutes.

"I am a backend developer with close to two years of experience, currently at Zensar working on banking systems for C. Hoare & Co, a private bank in the UK.

My work is Java and Spring Boot with Spring Data JPA and MySQL, and I have built two platforms there: a transaction recall system and a business card management system. Both were REST APIs with role-based access control, and both had performance problems when I picked them up — I took report generation from 15 seconds to 6, and the card APIs from 2 to 3 seconds down to under a second, mostly by removing N+1 queries and indexing properly.

The part I enjoyed most was moving work off the request thread. I replaced synchronous email sending with RabbitMQ so bulk approvals stopped timing out, and used Kafka to publish transaction state changes to compliance and reporting consumers.

I have also built a microservices project of my own with Eureka and an API Gateway, which is why this role interested me — it is Java, Spring Boot and microservices, which is exactly the direction I have been moving in."

Practise this twice today and twice on the morning of the 5th. Time it.

Your four stories, prepared properly

For each: **what it was** → **what you decided** → **what was hard** → **what you would change**.
The fourth is what makes an answer sound senior.

Story 1 ● — The report that took 15 seconds (*lead with this*)

"Report generation on the recall platform was taking about 15 seconds, and users were complaining. My first assumption was that the query itself was heavy, but I turned on SQL logging and saw the same query repeating once per row — a classic N+1, because the report was walking each transaction's related entities.

I replaced it with a projection query that joined and selected only the columns the report needed, so it stopped loading full entities. Then I looked at the execution plans, added indexes on the columns used in the WHERE and JOIN clauses, and rewrote a couple of JOINS that were forcing full scans. That took it from 15 seconds to 6.

What I took from it is to measure first. My initial guess was wrong, and if I had acted on it I would have spent a day tuning a query that was not the problem."

Follow-ups to be ready for: *What is N+1?* (file 04) *How did you know which index to add?* (**EXPLAIN** , file 08) *Why 6 seconds and not 1?* — be honest: some of it was the sheer data volume and the report's scope, and the next step would have been caching or pre-aggregating.

Story 2 ● — The APIs that timed out on bulk approvals

"On the business card system, approvals were sending emails synchronously inside the request. A single approval was slow, and bulk approvals were timing out — which was worse than slow, because the approval had actually succeeded but the user saw a failure.

I moved the email to a RabbitMQ message and a separate consumer. The API went back to responding in milliseconds, and a mail server problem stopped being an API problem — the messages just wait in the queue.

Separately the same APIs had N+1 issues, so I fixed those and set the right associations to lazy, which brought response times from 2 to 3 seconds down to under a second."

The reasoning to make explicit: the email is not something the user is waiting for, so it does not belong in the request.

Story 3 — Kafka for transaction state changes

Use this when asked about architecture or about Kafka. The key point is *why* Kafka and not a queue — the full answer is in file 06, Q1.

"Transaction state changes needed to reach more than one consumer — compliance logging and report generation, with more likely later. With a queue the first consumer takes the message and the others never see it. With Kafka each consumer group reads the same events independently at its own pace, the events are retained so a consumer that was down can catch up, and compliance can replay history if it needs to rebuild."

Story 4 — A production incident

Your resume says you resolved production incidents by analysing execution plans and logs. Have one concrete example ready, in STAR form: what broke, how you found it, what you did, what happened, and what changed afterwards so it would not recur.

The part that matters most is the last one. "We added an alert / added an index / added a test so it would be caught earlier" turns a firefighting story into a process-improvement story — which is exactly the JD's *"assess current processes, identify improvement areas"*.

Your Classifieds project

Covered in file 05, Q7. Keep it brief unless they ask — your professional work is stronger. Its value is showing that you have designed a microservices system end to end rather than only worked inside one service.

●● Section 5 — The two hard questions your resume creates

Both of these will come up. Neither is a problem. Both are a problem if you are unprepared.

1. "You were a Talent Acquisition Executive. Why the switch to development?"

This is certain. Answer it directly, without apology, and turn it into a strength.

The shape that works:

"I finished my B.Tech in Computer Science in 2022, so development was always the direction I wanted. I did the JSPIDERS Java full stack programme through 2023 to build the skills properly.

The recruitment role at Artifint was a technology recruitment role — I was screening and coordinating interviews for technical positions, so I was around the work I wanted to be doing. [Add the real reason it happened — it was the opportunity available at the time / for family reasons / while I finished the training.]

I moved into engineering at Zensar in December 2024 and have been building backend systems since. Honestly, the recruitment year was more useful than it sounds — I spent it talking to engineers and clients every day, and I am comfortable in requirement conversations and client calls in a way that is not always true of people who came straight through."

Why that last paragraph matters: read the JD again. *"Work with clients", "seek clarifications", "collaborate", "create requirement specifications"*. Infosys is a consulting business. Communication is a genuine differentiator there, and you have evidence for it that most candidates do not.

⚠ **Fill in the real reason yourself. Do not invent one.** A career change is completely normal and no interviewer will hold it against you. Evasiveness is what they react to.

2. "So how much experience do you have exactly?"

Look at your own resume carefully. The summary says **"2+ years building event-driven backend systems"**, but your engineering role runs from **December 2024**, which is about **1 year 9 months** as of early September 2026.

That gap between the claim and the dates is the kind of thing an interviewer notices, and **Infosys runs formal background verification** where dates are checked against payslips and employment letters. Your offered band and salary are set from that number.

What to do before the 5th — pick one:

- **Best:** change the summary to something accurate — *"Java developer with close to 2 years"*, or simply *"Java developer building event-driven backend systems..."* with no number, and let the dates speak.
- **Or:** if you are counting total professional experience including the Artifint role, make that explicit rather than attaching "2+ years" to backend engineering.

In the interview, always answer with the dates, not a rounded number:

"I have been in my engineering role since December 2024, so about a year and nine months, plus a year in technical recruitment before that."

That is straightforward, it is verifiable, and it removes the issue entirely. **Never inflate a date or a duration.** It is the single easiest thing to get caught on at background verification, and being caught costs the offer after you have already resigned somewhere.

Questions about your numbers

If they ask how you measured 15s → 6s, or 2-3s → under 1s, be ready with: application timings and logs before and after, plus the query counts from SQL logging. If the numbers came from someone else's measurements or a monitoring dashboard, say so — attributing accurately is better than being vague.

✓ Before the interview

1. Say the "tell me about yourself" script out loud, twice. Time it.
2. Write down the **real one-sentence reason** for the recruitment year, and say it until it sounds calm.
3. Decide how you will state your experience duration, and make the resume match.
4. Be able to say every number without hesitating: **15→6, 2-3→under 1, 3 services, 200+ profiles**.
5. Have one production-incident story with a **what changed afterwards** ending.

12 · HR, Managerial and the Offer Conversation

Time needed: 60 minutes. Say every answer out loud.

Infosys usually runs a separate managerial or HR round, and it is not a formality. People clear the technical round and lose the offer here.

1. "Why Infosys?"

Do not say "it is a big company" or "for growth". Say something only someone who read the posting could say.

"Three reasons.

The role is Java, Spring Boot and microservices, which is exactly what I have been doing and where I want to go deeper. That is a straightforward fit.

Second, I have banking domain experience from working on a UK private bank's systems, and Infosys works across financial services. I would rather build on a domain I already understand than start from nothing.

Third, the job description talks a lot about working with clients — requirement analysis, seeking clarifications, proposing alternatives. I have done that at Zensar with a client team in another country, and before that I spent a year in technical recruitment, so I am comfortable in those conversations. It is the part of the work I actually enjoy, and it seems central to how Infosys operates."

That third point is your real differentiator. Use it.

2. "Why are you leaving Zensar?"

Never criticise your current employer. Frame it as moving toward something.

"I have had good exposure at Zensar — I have owned features end to end on two banking platforms and worked across the full SDLC. What I am looking for now is more depth in microservices architecture rather than mostly working within one application, and more direct client-facing work. This role is explicitly microservices and consulting oriented, which is why I applied."

If asked about compensation, it is acceptable to say it is *a* factor, but never the first one you mention.

3. The JD's own language — prepare these four

The posting is full of consulting phrases. Each one is a question waiting to be asked.

"How do you handle unclear or incomplete requirements?" *(the JD's "problem definition" and "seek clarifications")*

"I do not start building on an assumption. I write down what I have understood, including the gaps, and take it back with specific questions rather than 'please clarify'. On the recall workflow there were states where it was not defined who could act, so I listed the cases, proposed a default for each, and asked them to confirm. That is much faster to answer than an open question, and it gets a decision recorded."

"How do you estimate effort?" *(the JD's "effort estimation")*

"I break the work into pieces small enough to reason about — API changes, data model changes, testing, integration — and estimate those rather than the feature as a whole. I add time explicitly for testing and for integration issues, because those are what people forget. And I flag what the estimate assumes, so if an assumption turns out to be wrong the estimate can be revisited rather than just being missed."

"How do you learn a new technology?" *(the JD's "research, literature surveys, POCs")*

"Official documentation first, because tutorials teach you one path and the docs tell you the model. Then I build a small proof of concept, because reading does not surface the problems. When I picked up Kafka I built a small producer and consumer before touching the real system, which is where I actually understood consumer groups and offsets. Then I check how it is used in the existing codebase, since conventions there matter more than my preferences."

"How do you approach diagnosing a production issue?" *(the JD's "diagnose the root cause")*

Use your real process from file 10, Design 5: reproduce if possible, check logs and timings, form a hypothesis, verify it before fixing, fix, then add something that would have caught it earlier. Finish with a concrete example.

4. Standard behavioural questions — use STAR

Situation → **T**ask → **A**ction → **R**esult. Under 90 seconds. Always end on a result.

"Tell me about a challenging technical problem." → The 15s → 6s story (file 11).

"A time you disagreed with someone." → Best if it is a technical disagreement you resolved with evidence rather than opinion. The pattern that works: you raised the concern, then brought data or a working alternative instead of just objecting.

"A time you made a mistake." → Have one. Refusing to answer this looks worse than the mistake. Pick something real, own it plainly, and finish with what you changed. A missed edge case caught in QA, with "and I added a test for that state transition afterwards", is a good shape.

"**How do you handle pressure or a tight deadline?**" → Scope, communicate early, do not silently slip. You worked in SIT and UAT cycles, so you know what happens when something half-finished reaches testing.

"**What is your weakness?**" → Real, bounded, with the fix.

"Coming from a non-linear route into development, I was initially slower at reading large unfamiliar codebases than people who had been doing it longer. I have got much better by tracing one request end to end through the code before changing anything, rather than jumping straight to the file I think is responsible."

Never "I am a perfectionist" or "I work too hard".

"**Where do you see yourself in 3 years?**"

"Owning a service or a module end to end, and being the person on the team who is trusted with performance and data-layer problems. Longer term I would like to move toward solution design, which is part of why the consulting side of this role appeals."

"**How do you work in a team?**" → Mention code reviews (you do them), QA coordination (on your resume), and being specific in reviews rather than vague.

5. Logistics questions — answer instantly

Question	How to answer
Notice period	State it exactly. If it is 60 or 90 days, say so, and mention whether buyout is possible. Do not guess.
Willing to relocate?	Infosys posts across Hyderabad, Bangalore, Pune, Chennai, Mysore. Decide your real answer now . If you are flexible, say so clearly — it is a genuine advantage. If not, say which locations work rather than a flat no.
Work from office?	Infosys is largely back in office. Answer directly.
Shifts / on-call?	Client projects sometimes need overlap with UK or US hours. You already work with a UK client, so say that — you know what it involves.
Do you have other offers?	Honest and brief. Do not invent leverage; it usually backfires.

⚠ If any of these are genuinely a problem, say so **now**, politely. Accepting and renegotiating later damages the relationship and your reference.

6. Salary

Know your numbers before the call: your current fixed CTC, your current total CTC, and the number you want. Do not do arithmetic live.

When asked for expectations:

"My current CTC is [X]. Based on my experience with Java, Spring Boot, microservices and messaging, and my banking domain background, I am looking for [Y]. But I am interested in the role and the project, so if the fit is right I am open to discussing it."

Rules:

- **Never lie about your current CTC.** Infosys verifies with payslips and Form 16. Being caught costs the offer, usually after you have resigned elsewhere.
- **Ask for the breakup** at offer stage: fixed, variable, retention bonus, and what the in-hand figure is. Large companies often put a meaningful share in variable pay.
- **Ask about the band or level.** Infosys has defined levels, and the band affects your future increments more than the starting number does.
- If they ask first, it is completely normal to ask **"what range have you budgeted for this role and level?"**

7. • Questions YOU ask them

Have four ready, ask two or three. Ending with "no, I'm good" reads as no interest.

Technical:

1. "Which domain would this role sit in — is it financial services, given the skills?"
2. "Is the project greenfield microservices, or modernising an existing monolith?"
3. "What does the current stack look like — Spring Boot version, cloud platform, CI/CD?"
4. "How large is the team, and would I own a service end to end?"

Role and growth:

5. "What would a successful first three months look like?"
6. "Is there scope to move toward design and client-facing work over time?"
7. "How does Infosys handle certification and upskilling for someone at this level?"

The one that makes you memorable, for the managerial round:

"What is the biggest technical problem the team is dealing with right now that this hire would help with?"

Then **listen**, and connect your answer to it. That is how you close.

8. On the day

- Two printed copies of your resume, and a copy open on your laptop.
- **Know your own resume cold.** Everything on it is fair game — Kafka, RabbitMQ, Eureka, JWT, Hibernate lazy loading, MongoDB, execution plans. If you touched something briefly, say so honestly rather than overstating.
- Carry your ID, and have your employment and education documents ready — Infosys onboarding is document heavy.

- If it is online: join five minutes early, test camera and microphone, sit somewhere quiet with a plain background, and keep an offline copy of your notes in case the network drops.
- Dress smart. Slightly overdressed beats underdressed.

| 9. If you get stuck

- **Do not freeze — narrate.** "Let me think about that for a second" is a complete and acceptable sentence.
- **Do not bluff.** "I have not used that directly. My understanding is X — is that the direction you mean?" has never cost anyone an offer. Bluffing has, because the next question exposes it.
- **Ask what they are getting at.** "Do you mean at the database level or in the application?"
- If you fumble one question, let it go. People lose interviews by replaying question 4 while answering question 7.

13 · Final Cheat Sheet — INTERVIEW MORNING ONLY

45 minutes. Nothing else. No new material.

If a line here does not ring a bell, open that file for three minutes and come back.

| The 8 answers that carry this interview

1. **HashMap:** "`hashCode()` , then a spread of `h ^ (h >>> 16)` , then bucket index `(n-1) & hash` . Collisions chain in a linked list, and since Java 8 a bucket over 8 entries becomes a red-black tree. Default capacity 16, load factor 0.75, so it resizes and rehashes at 12."
 2. **equals/hashCode:** "Equal objects must have equal hash codes. Override `equals` without `hashCode` and the map looks in the wrong bucket and never finds the entry."
 3. **N+1:** "One query for the list, then one per row for the relation. Fix with JOIN FETCH, `@EntityGraph` , batch fetching, or a DTO projection. That is what took my report from 15 seconds to 6."
 4. **@Transactional internal call:** "It works through a proxy. An internal `this.method()` call bypasses the proxy, so no transaction starts."
 5. **Constructor injection:** "Fields can be final, dependencies cannot be missing, and I can test with `new Service(mock)` without a Spring context."
 6. **Kafka vs RabbitMQ:** "RabbitMQ is a postbox — delivered to a worker, then gone. Kafka is a ledger — events are retained and many consumer groups read independently. I used RabbitMQ for email tasks and Kafka for state-change events with multiple consumers."
 7. **Circuit breaker:** "Closed, open, half-open. Failing fast stops the caller exhausting its threads and gives the failing service room to recover."
 8. **Index cost:** "Faster reads, but every insert and update maintains every index, so I add them deliberately, not by default."
-

| Core Java

- **Overloading** = compile time, different parameters. **Overriding** = runtime, same signature, subclass. You cannot override a static method — it is hidden.
- **Abstract class** = shared state and code, one only. **Interface** = capability contract, many.
- **ArrayList** = array, O(1) get. **LinkedList** = nodes, O(n) get. Default to ArrayList.
- **HashMap** not thread safe · **Hashtable** locks the whole map, legacy · **ConcurrentHashMap** locks per bucket.
- **String immutable** because of the string pool, security, thread safety, cached hash.
- `==` compares references, `.equals()` compares values. `new String("a") != "a"` .

- **Checked** = compiler forces handling. **Unchecked** extends `RuntimeException`.
- `finally` always runs except on `System.exit()`. Never `return` from `finally`.
- **Fail-fast** throws `ConcurrentModificationException`; remove via `iterator.remove()`.

Java 8

- **Intermediate** ops are lazy (`filter`, `map`, `sorted`); **terminal** ops trigger (`collect`, `reduce`, `forEach`, `findFirst`).
- `map` = one to one. `flatMap` = one to many, then flattened.
- `Predicate` → filter · `Function` → map · `Consumer` → forEach · `Supplier` → lazy value.
- `groupingBy(X, counting())` and `groupingBy(X, summingDouble(Y))`.
- **Optional**: use `map` / `orElse` / `orElseThrow`, not `isPresent` + `get`.
`orElse` always evaluates; `orElseGet` is lazy.

Spring Boot

- **IoC** = the framework controls creation. **DI** = how it hands dependencies in.
- Beans are **singleton by default**, so keep them **stateless** — a mutable instance field is a concurrency bug.
- `@Repository` also **translates database exceptions** into Spring's hierarchy.
- `@RestControllerAdvice` + `@ExceptionHandler` = one consistent error shape. Log the stack trace, return a generic message.
- **Auto-configuration** works through `@ConditionalOnClass` and `@ConditionalOnMissingBean` — your own bean always wins.
- `@WebMvcTest` = slice, fast · `@SpringBootTest` = full context · `@MockBean` = replace a bean.

JPA and Hibernate

- **JPA** = spec · **Hibernate** = implementation · **Spring Data JPA** = the layer on top.
- Defaults: `@ManyToOne` and `@OneToOne` are **EAGER**; `@OneToMany` and `@ManyToMany` are **LAZY**. Set `ManyToOne` to `LAZY` explicitly.
- `@Transactional` rolls back on **unchecked** exceptions only, unless `rollbackFor`.
- **Dirty checking** means a managed entity needs no `save()`.
- First-level cache is always on, per transaction. Second-level is opt-in and shared.
- Use `@Enumerated(EnumType.STRING)`, never the `ORDINAL` default.
- Money is **BigDecimal**, never `double`.

Microservices

- Gains: independent deploy, independent scale, fault isolation. **Costs: network failure, no distributed transaction, harder debugging**. Say the costs.

- **Eureka** = discovery · **API Gateway** = single entry, auth and routing in one place · **Config Server** = externalised config.
- **Saga** = local transactions plus **compensating** transactions. Eventual consistency.
- Only retry **idempotent** operations. Always set a **timeout**.
- Split by **business capability**, not technical layer.
- Observability: centralised logs, **correlation ID** tracing, metrics.

Messaging

- Kafka: **topic** → **partitions** → **offsets**; order is guaranteed **within a partition**, so key by transaction or account ID.
- **Same consumer group** = share the partitions. **Different groups** = each gets everything.
- **At-least-once** is the normal default, so consumers must be **idempotent**.
- RabbitMQ: producer → **exchange** → queue. Direct, topic, fanout. **DLQ** after N failed attempts.

Security

- **JWT** = header.payload.signature. **Signed, not encrypted** — anyone can read the payload, so put nothing sensitive in it.
- Stateless, so any instance validates it — that is why it suits microservices.
- Cannot easily revoke: short-lived access token plus refresh token, or a blocklist.
- CSRF is off for a stateless API because the browser does not auto-attach an `Authorization` header.
- **BCrypt** for passwords — slow and salted by design.
- `hasRole("ADMIN")` looks for the authority `ROLE_ADMIN`.

SQL

- **Left-most prefix rule:** an index on `(status, created_at)` does not serve `created_at` alone.
- Index killers: a **function on the column**, and a **leading %** in LIKE.
- **EXPLAIN**: `type: ALL` means full scan · check `key`, `rows`, and `Extra`.
- **WHERE** filters rows **before** grouping; **HAVING** filters groups **after**.
- **UNION** deduplicates and sorts; **UNION ALL** does not and is faster.
- Customers with no transactions = `LEFT JOIN ... WHERE t.id IS NULL`.
- Deep **OFFSET** is slow — prefer **keyset pagination**.

Coding round

- Talk while you type. Ask about edge cases **first**. State the complexity at the end.
- `LinkedHashMap` for first-non-repeating. `seen.add(n)` returns false for a duplicate.

- Two sum = HashMap, O(n), trading space for time. Mention the brute force first.
 - `Integer` cache is -128 to 127, so `128 == 128` is **false** for boxed Integers.
-

| Your stories — the numbers

15s → 6s report: N+1 found via SQL logging, fixed with a projection query, indexes, rewritten JOINS. *"My first guess was wrong, which is why I measure first."*

2-3s → under 1s APIs: N+1 plus lazy loading plus indexing.

RabbitMQ: email off the request thread, bulk approvals stopped timing out.

Kafka: state changes to multiple consumer groups, retained and replayable.

RBAC: requester, approver, admin — enforced server side, because a UI check only hides the button.

Classifieds: 3 services, own databases, Eureka, API Gateway, JWT, global exception handling.

| The three questions you must not fumble

"Tell me about yourself" → Now → What you built → Why this role. Under 2 minutes.

"Why did you move from recruitment to development?" → B.Tech in CS, JSPIDERS training, the recruitment year was technical recruitment, moved to engineering Dec 2024.

Then: *"That year made me comfortable in client and requirement conversations, which is a lot of what this role is."*

"How much experience do you have?" → **Answer with dates, never a rounded number.**

"Engineering since December 2024, so about a year and nine months, plus a year in technical recruitment before that."

| Questions to ask them

"Which domain would this role sit in?" · "Greenfield microservices or modernising a monolith?" · "What would a successful first three months look like?" ·

"What is the biggest technical problem this hire would help with?"

| The last five minutes

- You have real production numbers and banking domain experience. Most candidates for this role have neither.
- **If you do not know something: say so, then reason out loud.** That has never cost anyone an offer. Bluffing has.
- Give the **why**, not just the **what**. "HashMap stores key-value pairs" is a textbook answer; "O(1) because it hashes to a bucket instead of scanning" is an engineer's.
- Sit up. Breathe out slowly. Smile before you speak — it changes your voice.

Go get it.